

LECTURE SHEET

EXERCISE-I

(Classification of elements, Mendeleev's periodic table, Modern periodic law, Modern periodic table, Classification of elements into blocks, Classification of elements into types, periodicity, atomic size, atomic radii, ionic radii in groups and periods)

LEVEL-I (MAIN)

Straight Objective Type Questions

- The element 'Sc' is known long back as
 - 1) eka-aluminium
 - 2) eka-boron
 - 3) eka-silicon
 - 4) eka-mercury
- By taking chemical properties into consideration, the atomic weights of the following elements were corrected
 - 1) Te & I
 - 2) Ar & K
 - 3) Co & Ni
 - 4) Be & In

3. Anomalous pair among the following is
1) Boron - Silicon
2) Beryllium - Indium
3) Aluminium - Gallium
4) Cobalt - Nickel
4. The general electronic configuration of elements of carbon family
1) $ns^2 np^4$
2) $ns^2 np^3$
3) $ns^2 np^1$
4) $ns^2 np^2$
5. Outer shell 'octet' configuration is observed for the elements of the group
1) 2
2) 8
3) 18
4) 32
6. The starting element of fifth period is
1) K
2) Rb
3) Kr
4) Xe
7. Element with atomic number 15 and mass number 31 is present in
1) group 5 and period 4
2) group 5 and period 3
3) group 15 and period 3
4) group 15 and period 4
8. In the periodic table, the elements are arranged in the periods following the
1) Hund's rule of maximum multiplicity
2) Pauli's exclusion principle
3) Aufbau principle
4) Both (1) and (2)
9. The general electronic configuration of d-block elements is
1) $ns^{1-2} (n-1)d^{1-10}$
2) $ns^2(n-1)d^1(n-2)f^{1-14}$
3) $ns^{1-2} (n-1)d^{1-9}$
4) $ns^{1-2} np^6 (n-1)d^{1-10}$
10. Identify the correctly matched set among the following
1) Scandium-d-block-representative element
2) Lanthanum-d-block-innertransition element
3) Cerium-f-block-transition element
4) Actinium-d-block-transition element
11. The representative elements get the nearest inert gas configuration by
1) losing electrons
2) gaining electrons
3) sharing electrons
4) losing or gaining or sharing electrons
12. In transition elements, the shells that are incompletely filled
1) Ultimate shell only
2) Penultimate shell only
3) Both ultimate and penultimate shells
4) Outermost three shells
13. Elements of p-block are
1) Only non-metals
2) Only metalloids
3) Metalloids and non-metals
4) Metalloids, non-metals and metals
14. The following ion is coloured in aqueous solution
1) Ca^{2+}
2) Sc^{3+}
3) Zn^{2+}
4) Ti^{3+}
15. Which of the following configuration corresponds to an inert gas?
1) $1s^2 2s^2 2p^5$
2) $1s^2 2s^2 2p^6$
3) $1s^2 2s^1$
4) $1s^2 2s^2 2p^6 3s^1$

16. The rare gas that is most abundant in the atmosphere is
1) He 2) Ne 3) Ar 4) Kr
17. In lanthanides, the differentiating electron enters into
1) d - subshell 2) f - subshell 3) p - subshell 4) s - subshell
18. Which is not a transition metal?
1) Ag 2) Pb 3) Cr 4) Pt
19. The general electronic configuration $(n-1)d^3ns^2$ indicates that the particular element belong to the group
1) VB 2) VA 3) IVB 4) IIB
20. The electronic configuration of chromium is
1) $[Ar] 4d^4 4s^2$ 2) $[Ar] 4d^5 4s^1$ 3) $[Ar] 3d^4 3s^2$ 4) $[Ar] 3d^5 4s^1$
21. Which of the following pairs of ions have the same electronic configuration
1) Cr^{+3}, Fe^{+3} 2) Sc^{+3}, Cr^{+3} 3) Fe^{+3}, Co^{+3} 4) Fe^{+3}, Mn^{+2}
22. The electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^4$ represents
1) Oxygen 2) Magnesium 3) Calcium 4) Sulphur
23. Covalent bond length of chlorine molecule is $1.98\text{\AA}$. Then covalent radius of chlorine is
1) $1.98\text{\AA}$ 2) $1.7\text{\AA}$ 3) $2.05\text{\AA}$ 4) $0.99\text{\AA}$
24. Van der waal's radius is used for
1) Molecular substances in gaseous state only
2) Molecular substances in solid state only
3) Molecular substances in liquid state only
4) Molecular substances in any state
25. If the atomic radius of non-metal bromine is $1.14\text{\AA}$, its covalent radius is
1) $1.14\text{\AA}$ 2) $1.12\text{\AA}$ 3) $1.16\text{\AA}$ 4) $0.57\text{\AA}$
26. The covalent and van der Waals radii of chlorine respectively are
1) $1.80\text{\AA}$ & $0.99\text{\AA}$ 2) $0.99\text{\AA}$ & $1.80\text{\AA}$
3) $1.80\text{\AA}$ & $1.80\text{\AA}$ 4) $0.99\text{\AA}$ & $0.99\text{\AA}$
27. In the isoelectronic species the ionic radii ($\text{\AA}^0$) of N^{3-} , O^{2-} and F^- are respectively given by
1) 1.36, 1.71, 1.40 2) 1.36, 1.40, 1.71
3) 1.71, 1.36, 1.40 4) 1.71, 1.40, 1.36
28. Correct statement among the following is
1) Covalent radius is 40% more than Van der waals radius
2) Van der waals radius is less than covalent radius
3) Van der waal's radius is 40% more than covalent radius
4) Radii cannot be compared

LEVEL-II (ADVANCED)

Straight Objective Type Questions

- Atomic number of Ag is 47. In the same group the atomic number of elements placed above and below Ag will be
a) 37, 67 b) 29, 79 c) 39, 69 d) 18, 28
- Which of the following statement is not correct regarding hydrogen atom?
a) It resembles halogens in some properties
b) It resembles alkali metals in some properties
c) It can be placed in VII group of periodic table
d) It can not be placed in first group of periodic table
- Which of the following statement is wrong for the transition elements?
a) Last electron enters in $(n-1)d$ subshell
b) Transition elements are placed from 3rd to 6th period
c) Exhibits variable valency
d) General electronic configuration is $(n-1)d^{1-10}ns^{1-2}$
- Which of the following statement is/are incorrect about the modern form of periodic table?
a) Third group of periodic table accommodates maximum number of elements
b) Due to presence of half filled and fully subshells in electronic configuration electronegativity of atom increases
c) The element of 13th group and 7th period will have atomic number 113
d) Diagonal relationship in 2nd and 3rd period element is found due to similar polarising power.
- Element in periodic table with electronic configuration as $[\text{Ar}]3d^5 4s^1$ is placed in
a) IA, s-block b) VIA, s-block c) VIB, s-block d) VIB, d-block
- The ground state electronic configurations of some elements, A, B, C, D and E (these symbols represent the some of the known elements given in the periodic table) are as follows.
P) $1s^2 2s^2 2p^6 3s^2 3p^2$
Q) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
R) $1s^2 2s^2 2p^6 3s^2 3p^1$
S) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$
T) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$

Match the electronic configurations of the elements with the properties given below and select the correct sequence by choosing the correct codes given.

- Element forms a cation which is isoelectronic with P^{3-} .
- Element which in its compounds can show a maximum oxidation state of +6 and that is coloured too.
- Element has largest atomic radius and highest first ionisation energy in the respective period.
- Element which has intermediate value of electronegativity and its oxide forms salts with strong acids and bases.

a) QRT P

b) QSTR

c) QRST

d) PQRS

7. Which of the following is the wrong statement?
 a) All the actinide element are radioactive
 b) Alkali and alkaline earth metals are s-block elements
 c) Chalcogens and halogens are p-block elements
 d) The first member of the lanthanide series is lanthanum
8. Which of the following statements related to the modern periodic table is incorrect?
 a) The p-block has 6 vertical columns, i.e., groups
 b) The d-block has 8 vertical columns
 c) Each block contains a number of columns equal to the number of electrons that can occupy that subshell
 d) The block indicates value of azimuthal quantum number (l) for the last subshell that received electrons in building up the electronic configuration
9. M^{3+} has electronic configuration as $[Ar]3d^{10}4s^2$, hence it lies in
 a) s-block b) p-block c) d-block d) f-block
10. Which of the following is true about the element $_{33}As$?
 a) It is the 5th period element b) It is p-block element
 c) It belongs to 16th group d) It is the member of VA group
11. Total number of elements in 5th period are :
 a) 8 b) 18 c) 32 d) 28
12. Which of the following order of radii is correct?
 a) $Li < Be < Mg$ b) $H^+ < Li^+ < H^-$ c) $O < F < Ne$ d) $Na^+ > F^- > O^{2-}$
13. Which of the following is not isoelectronic series?
 a) Cl^- , P^{3-} , Ar b) N^{3-} , Ne , Mg^{2+} c) B^{3+} , He , Li^+ d) N^{3-} , S^{2-} , Cl^-
14. Which of the following order of atomic/ionic radius is not correct?
 a) $I^- > I > I^+$ b) $Mg^{2+} > Na^+ > F^-$ c) $P^{5+} < P^{3+}$ d) $Li > Be > B$
15. Screening effect is not observed in :
 a) He^+ b) Li^{2+} c) Be^{3+} d) in all cases
16. Select correct statement about radius of an atom
 a) values of van der Waal's radii are larger than those of covalent radii because the van der waal's forces are much weaker than the forces operating between atoms in covalently bonded molecule
 b) the metallic radii are smaller than the van der waal's radii, since the bonding forces in the metallic crystal lattice are much stronger than the vander waal's forces.
 c) both are correct d) none is correct
17. Which of the following series of elements have nearly the same atomic radii?
 a) F, Cl, Br, I b) Na, K, Rb, Cs c) Li, Be, B, C d) Fe, Co, Ni, Cu
18. In a given energy level, the order of penetration effect of different orbitals is
 a) $f < d < p < s$ b) $s = p = d = f$ c) $s < p < d < f$ d) $p > s > d > f$
19. The lanthanides contraction refers to
 a) radius of the series b) valence electrons of the series
 c) the density of the series d) electronegativity of the series

20. Which represents alkali metals (i.e., I group metals) based on $(IE)_1$ and $(IE)_2$ values?

	$(IE)_1$	$(IE)_2$		$(IE)_1$	$(IE)_2$
a) X	100	110	b) Y	95	120
c) Z	195	500	d) M	200	250

21. In which of the following arrangements, the order is correct according to the property indicated against it :

- a) increasing IE_1 : Na < Al < Mg < Si
 b) increasing IE_1 : B < C < N < O
 c) increasing size : Cu^{2+} < Cu < Cu^+
 d) increasing IE_1 : Li < Na < K < Rb

22. Which of the following orders is not correct for the size ?

- a) Al^{3+} < Mg^{2+} < N^{3+} < F^-
 b) Te^{2-} > I^- > Cs^+ > Ba^{2+}
 c) Fe^{3+} < Fe^{2+} < Fe^{4+}
 d) Mg > Al > Si > P

23. Which one of the following sets of ions represents the collection of isoelectronic species?

- a) Na^+ , Mg^{2+} , Al^{3+} , F^-
 b) K^+ , Ca^{2+} , Sc^{3+} , Cl^-
 c) K^+ , Cl^- , Mg^{2+} , Sc^{3+}
 d) both (a) and (b)

24. Correct order of ionic radius is

- a) Ti^{4+} < Mn^{7+}
 b) Cl^- < Cl
 c) K^+ > Cl^-
 d) P^{3+} > P^{5+}

More than One correct answer Type Questions

25. Which of the following element(s) belongs to halogen group?

- a) $1s^2, 2s^2, 2p^6, 3s^2 3p^5$
 b) $1s^2, 2s^2, 2p^6, 3s^2 3p^6, 4s^2$
 c) $[Ar]3d^{10}, 4s^2 4p^5$
 d) $[Kr]4d^{10}, 5s^2 5p^5$

26. The element Ge belongs to

- a) Fourth period
 b) p-Block
 c) Group 14
 d) Inert gases

27. Which of the following species will be isoelectronic with Ne?

- a) N^{3-}
 b) Na^+
 c) Al^{3+}
 d) F^-

28. The classification of elements in the long form of the periodic table is based on

- a) Atomic weight
 b) Atomic number
 c) N/P ratio
 d) Electronic configuration

29. Which of the following pairs contain elements with similar atomic radii ?

- a) Co, Ni
 b) Zn, Mo
 c) Rh, Ir
 d) Hf, Ti

30. Select correct about element UuO.

- a) It belongs to 7th period
 b) It belongs to 18th group
 c) It is a p-block element
 d) It is an inert gas

31. Incorrect order of ionic size of elements:

- a) Mn^{7+} > Mn^{6+} > Mn^{4+}
 b) C^+ > C > C^-
 c) Fe^{3+} > Fe^{2+} > Fe
 d) P^{3-} > P > P^{3+}

Linked Comprehension Type Questions

Passage-I :

Numerous forms of the periodic table have been devised from time to time. A modern version which is most convenient and widely used is the long or extended form of periodic table. The aufbau principle and the electronic configuration of atoms provide a theoretical foundation for the periodic classification. The horizontal rows are called periods. There are altogether seven periods. The first period consists of 2 elements. The subsequent periods consist of 2, 8, 8, 18, 18 and 32 elements respectively. The seventh period is incomplete and like the sixth period would have maximum of 32 elements. Elements having similar outer electronic configurations in their atoms are grouped in vertical columns.

32. An element belongs to group 17 with atomic number is 17. What is the atomic number of the element belonging to same group and present in fifth period?
 a) 25 b) 33 c) 35 d) 53
33. What is the position of the element in the periodic table satisfying the electronic configuration $(n-1)d^1 ns^2$ for $n = 4$?
 a) 3rd period and 3rd group b) 4th period and 3rd group
 c) 3rd period and 2nd group d) 4th period and 2nd group
34. Elements A, B, C, D and E have the following electronic configuration:
 A) $1s^2, 2s^2 2p^1$ B) $1s^2, 2s^2 2p^6, 3s^2 3p^1$ C) $1s^2, 2s^2 2p^6, 3s^2 3p^3$
 D) $1s^2, 2s^2 2p^6, 3s^2 3p^5$ E) $1s^2, 2s^2 2p^6, 3s^2 3p^6$
 Which among these will belong to same group in the periodic table ?
 a) A and C b) A and D c) A and B d) A and E

Passage-II :

Numerous forms of the periodic table have been revised from time to time. A modern version, which is most convenient and widely used is the long or extended form of periodic table. The aufbau principle (electrons are filled in the progressive order of their increasing energy i.e. by $n + 1$ rule) and the electronic configuration of atom provides a theoretical foundation for the periodic classification. The horizontal rows are called periods. There are altogether seven periods. The first period consists of 2 elements. The subsequent periods consist of 2, 8, 8, 18, 18 and 32 elements respectively. The seventh period is incomplete and like the sixth period would have maximum of 32 elements.

35. The element with atomic number 56 is likely to have the same outer shell configuration as the element with atomic number :
 a) 12 b) 18 c) 14 d) 16
36. Elements a, b, c, d and e have the following electronic configurations.
 A) $1s^2, 2s^2 2p^1$ B) $1s^2, 2s^2 2p^6, 3s^2 3p^1$ C) $1s^2, 2s^2 2p^6, 3s^2 3p^3$ D) $1s^2, 2s^2 2p^6, 3s^2 3p^5$
 E) $1s^2, 2s^2 2p^6, 3s^2 3p^6$
 Which among these will belong to the same group in periodic table.
 a) A and C b) A and B c) A and D d) D and E
37. If aufbau rule is not followed, Ca-20 will be placed in _____ block.
 a) s- b) p- c) d- d) f-

Matrix Matching Type Questions

38. Match the atomic number in Column-I, with groups in Column-II.

Column-I

- A) 55
B) 45
C) 81
D) 64

Column-II

- P) VIII
Q) IIIB
R) IIIA
S) IA

39. **Column-I**
(Atom)

- A) With atomic number = 29
B) Having electronic configuration $[\text{Kr}]5s^2 4d^{10}$
C) Which form ion with $3s^2, 3p^6, 3d^{10}$ configuration of outermost shell after the removal of 3 electrons
D) In which last entered electron have $n=4, l=0, m=0, s=\pm 1/2$ set of quantum number

Column-II

(IUPAC group no. in periodic table)

- P) 1
Q) 11
R) 2
S) 12
T) 13

Integer Type Questions

40. An element X belongs to group 14 and 2nd period of the periodic table. Its atomic number will be
41. Atomic numbers of some elements are 31, 34, 49, 50, 53, 81, 84, and 85. Number of elements in 5th period is ____
42. In a period, the element with largest atomic volume belongs to which of the groups ?
43. Find out the sum of all the digits of atomic number of the element "Uuq".
(If your answer is greater than 9 then add all the digits.) For example, Ans. 12 then fill $1+2=3$,
Ans. 123 then fill $1+2+3=6$ and Ans. 20 then fill $2+0=2$)
44. For an element, the spin only magnetic moment is 4.90 B.M. Hence the number of unpaired electrons in the valence shell of this element is (are):

EXERCISE-II

(Ionisation potential, Factors affecting ionisation potential, Ionisation enthalpy in groups and periods, Electron gain enthalpy (electron affinity) in groups and periods, Electronegativity, EN scales, EN values in groups and periods, Valency and oxidation states, Electro positivity, metallic properties, Acidic and basic strength of oxides and oxy acids and hydrides, Chemical reactivity of elements and diagonal relationship)

LEVEL-I (MAIN)**Straight Objective Type Questions**

1. Which of the following process refers to ionisation potential?

- 1) $X_{(s)} \rightarrow X^+_{(g)} + e^-$
3) $X_{(g)} \rightarrow X^+_{(g)} + e^-$

- 2) $X_{(g)} + aq \rightarrow X^+_{(aq)} + e^-$
4) $X_{(g)} + e^- \rightarrow X^-_{(g)}$

2. When the screening effect increases, ionisation energy
1) Decreases
2) Increases
3) First increases and then decreases
4) Remains constant
3. With an increase in the extent of penetration of valence electrons, ionisation energy
1) Decreases
2) Increases
3) Remains constant
4) Both are not related
4. The group of elements with highest second ionisation energy is
1) IIA group
2) Zero group
3) VIIA group
4) IA group
5. How many ionisation energies can carbon have? Electronic configuration of carbon in Ground state is $1s^2 2s^2 2p_x^1 2p_y^1$ and is in Excited state $1s^2 2s^1 2p_x^1 2p_y^1 2p_z^1$.
1) 1
2) 2
3) 4
4) 6
6. First ionisation potential of magnesium is greater than that of aluminium because
1) Aluminium atom is very large when compared to Mg
2) Aluminium has a stable electronic configuration
3) Magnesium has a stable electronic configuration
4) The electron affinity of Magnesium is positive (energy is absorbed)
7. Electrons with the highest penetrating power are
1) p-electrons
2) s-electrons
3) d-electrons
4) f-electrons
8. If the Ionisation potential (I.P.) of Na is 5.48 eV. The I.P. of K will be
1) 4.34 eV
2) 5.68 eV
3) 10.88 eV
4) 5.48 eV
9. A sudden jump between the values of second and third ionisation energies of an element is associated with configuration
1) $1s^2 2s^2 2p^6 3s^1$
2) $1s^2 2s^2 2p^6 3s^2 3p^1$
3) $1s^2 2s^2 2p^6 3s^2 3p^2$
4) $1s^2 2s^2 2p^6 3s^2$
10. Amongst the following elements (whose electronic configurations are given below), the one having the highest first ionization energy is
1) $[\text{Ne}]3s^2 3p^1$
2) $[\text{Ne}]3s^2 3p^3$
3) $[\text{Ne}]3s^2 3p^2$
4) $[\text{Ar}]3s^{10} 4s^2 4p^2$
11. The first ionization energy of Lithium will be
1) Greater than Be
2) Less than Be
3) Equal to that of Na
4) Equal to that of F
12. Which of the following is true regarding inert pair effect
1) due to poor shielding of p orbital
2) due to poor shielding of d & f orbital
3) due to poor shielding of s orbital
4) due to poor shielding of s & p orbital
13. Generally the ionisation potential in a period increases, but there are some exceptions. The one which is not an exception is
1) Be & B
2) N & O
3) Mg & Al
4) Na & Mg
14. Element that has the highest first ionisation energy among the following is
1) Ca
2) Mg
3) Al
4) Si

15. The ionization potential of nitrogen is more than that of oxygen because of
- 1) the greater attraction of the electrons by the nucleus
 - 2) the extra stability of the half filled p - orbitals
 - 3) the smaller size of nitrogen
 - 4) more penetration effect
16. First and second ionization energies of magnesium are 7.646 and 15.035 eV respectively. The amount of energy in kJ needed to convert all the atoms of magnesium into Mg^{+2} ions present in 12 mg of magnesium vapours is [Given $1\text{eV} = 96.5\text{kJ mol}^{-1}$]
- 1) 1.1
 - 2) 1.5
 - 3) 2.0
 - 4) 0.5
17. Electron affinity is
- 1) Energy required to take out an electron from an isolated gaseous atom
 - 2) The tendency of an atom to attract an electron towards itself
 - 3) Energy absorbed when an electron is added to an isolated atom in gaseous state
 - 4) Energy released when an electron is added to an isolated atom in the gaseous state
18. The property of an element that is not determined directly but is obtained indirectly using Born-Haber cycle
- 1) Ionisation potential
 - 2) Electron affinity
 - 3) Electronegativity
 - 4) Metallic character
19. Electron affinity is measured in
- 1) No units
 - 2) kcal mol^{-1}
 - 3) kJ mol^{-1}
 - 4) Both (2) and (3)
20. Which of the following is an endothermic process?
- 1) First electron affinity of chlorine
 - 2) Second electron affinity of oxygen
 - 3) Formation of NaCl from gaseous ions
 - 4) Hydration of $MgCl_2$
21. In a period from left to right, electron affinity
- 1) Increases
 - 2) Decreases
 - 3) Remains constant
 - 4) First increases and then decreases
22. In which of the following process maximum energy is released
- 1) $O_{(g)} + e^- \rightarrow O^-_{(g)}$
 - 2) $O^-_{(g)} + e^- \rightarrow O^{2-}_{(g)}$
 - 3) $S_{(g)} + e^- \rightarrow S^-_{(g)}$
 - 4) $S^-_{(g)} + e^- \rightarrow S^{2-}_{(g)}$
23. Ionisation energy of F^- is equal in magnitude with the electron affinity of
- 1) F^-
 - 2) F
 - 3) F^+
 - 4) F^{2+}
24. When an electron is added, energy is absorbed in which of the following?
- 1) C
 - 2) N
 - 3) F
 - 4) O
25. Pauling's electronegativity is based on
- 1) Electron affinity
 - 2) Ionisation potential
 - 3) Both IP and EA
 - 4) Bond energies
26. Pauling's electronegativity values for elements are useful in predicting
- 1) Polarity of the molecules
 - 2) Position in the E.M.F. series
 - 3) Coordination numbers
 - 4) Dipole moments

LEVEL-II (ADVANCED)

Straight Objective Type Questions

- Which of the following has maximum ionisation potential?
a) B b) Be c) C d) Ne
- IP_1 and IP_2 of Mg are 178 and 348 kcal mole⁻¹. The energy required for the reaction, $Mg \rightarrow Mg^{2+} + 2e^-$ is :
a) +170 kcal b) +526 kcal c) -170 kcal d) -526 kcal
- Which of the following sequence regarding the first ionisation potential of coinage metal is correct?
a) Cu > Ag > Au b) Cu < Ag < Au c) Cu > Ag < Au d) Ag > Cu < Au
- Arrange the isotopes of hydrogen in correct order of ionisation potential:
a) D = T = P b) D > T > P c) D < T < P d) D < T > P
- Considering the elements F, Cl, O and N, the correct order of their electron affinity values is :
a) F > Cl > O > S b) F > O > Cl > S c) Cl > F > S > O d) O > F > S > Cl
- The correct order of negative electron gain enthalpy is
a) Be < B < C < N b) Be < N < B < C c) N < Be < C < B d) N < C < B < Be
- Which of the following will have the most negative electron gain enthalpy and which the least negative?
a) F, Cl b) Cl, F c) Cl, S d) Cl, P
- Fluorine has the highest electronegativity among the ns^2np^5 group on the pauling scale, but the electron affinity of fluorine is less than that of chlorine because:
a) the atomic number of fluorine is less than that of chlorine
b) fluorine being the first member of the family behaves in an unusual manner
c) chlorine can accommodate an electron better than fluorine by utilising its vacant 3d-orbital
d) small size, high electron density and an increased electron repulsion makes addition of an electron to fluorine less favourable than that in the case of chlorine in isolated stage
- The order of first electron affinity of O, S and Se is :
a) O > S > Se b) S > Se > O c) Se > O > S d) S > O > Se
- The correct order of electro negativity is
a) F > Cl < Br > I b) F > O > N > C c) S < O < Se < Te d) all
- The electro negativity values of C, N, O and F
a) Increase from carbon to fluorine
b) Decrease from carbon to fluorine
c) Increase upto oxygen and is minimum at fluorine
d) is minimum at nitrogen and then increase continuously
- Select the incorrect statement(s)
a) E(I) of nitrogen atom is more than IE(1) of oxygen atom
b) Electron affinity of oxygen is less than selenium atom
c) Electronegativity on pauling scale is 2.8 times greater than electronegativity on Mulliken scale
d) The Cr^{6+} is smaller than Cr^{3+} .

13. Which of the following order(s) is/are incorrect?
- $O^{2-} < F^- < Na^+ < Mg^{2+}$ Increasing $Z_{\text{effective}}$
 - $Mg^{2+} < Na^+ < O^{2-} < F^-$ Increasing size
 - $O^{2-} < F^- < Na^+ < Mg^{2+}$ Increasing ionisation energy
 - $O^{2-} < F^- < Na^+ < Mg^{2+}$ Increasing electron affinity
14. Fluorine is more electronegative than nitrogen. The best explanation is that
- The valence electrons in F are on the average, a little farther to the nucleus than in N.
 - The charge on a F nucleus is +9, while that on N nucleus is +7
 - The nitrogen has half filled valence shell electron configuration, ns^2np^3 where as fluorine has partially filled electron configuration ns^2np^5 .
 - The electronegativity decreases from top to bottom in each of the group as the effective nuclear charge remains constant.
15. Following the Mulliken scale, what parameters are required to evaluate electronegativity?
- only electronegativity
 - only electron affinity
 - electron affinity and ionization energy
 - ionic potential and electronegativity
16. A, B and C are oxides of elements X, Y and Z respectively. X, Y and Z are in the same period of the periodic table. A gives an aq. solution which turns blue litmus red. B reacts with both strong acids and strong alkalis. C gives an aq. solution which is strongly alkaline.
- Which of the following statement is/are true?
- All the three elements are metals
 - The electronegativities decreases from X to Y to Z
 - The atomic radius increases in the order $X < Y < Z$
 - X, Y and Z could be phosphorus, aluminium and sodium respectively
- I, II, III only correct
 - I, III only correct
 - II, IV only correct
 - II, III, IV only correct
17. The electronegativity values of the elements are useful in predicting:
- bond energy of a molecule
 - polarity of a molecule
 - nature of an oxide
 - all
18. The order of basic character of given oxides is
- $Na_2O > MgO > CuO > SiO_2$
 - $MgO > SiO_2 > CuO > Na_2O$
 - $SiO_2 > MgO > CuO > Na_2O$
 - $CuO > Na_2O > MgO > SiO_2$
19. Amphoteric behaviour is shown by the oxides of
- Al and Ca
 - Pb and Ba
 - Cr and Mg
 - Sn and Zn
20. An element X occurs in short period having configuration ns^2np^1 . The formula and nature of its oxide is
- XO_3 , basic
 - XO_3 , acidic
 - X_2O_3 , amphoteric
 - X_2O_3 , basic
21. Increasing order of acidic character is
- $SO_3 > N_2O_5 > CO_2 > SiO_2$
 - $SO_3 < N_2O_5 < CO_2 < SiO_2$
 - $SO_3 < N_2O_5 > CO_2 < SiO_2$
 - $SO_3 > N_2O_5 < CO_2 < SiO_2$

22. Which one of the following oxides is not neutral?
 a) CO b) OF₂ c) NO₂ d) both (b) and (c)
23. The incorrect order of acidic strength is:
 a) Cl₂O₇ > SO₂ > P₄O₁₀ b) CO₂ > N₂O₅ > SO₃
 c) Na₂O > MgO > Al₂O₃ d) both (b) and (c)
24. Identify the incorrect order of acidic strengths of CO₂, CuO, CaO, H₂O.
 a) CaO < CuO < CO₂ < H₂O b) H₂O < CuO < CaO < CO₂
 c) CaO < H₂O < CuO < CO₂ d) all of these
25. Thallium shows different oxidation states because
 a) of its high reactivity b) of inert pair of electrons
 c) of its amphoteric nature d) its is a transition metal
26. Which element exhibit inert pair effect?
 a) Pb b) Al c) Tl d) both a and c
27. In which of the following element +3 oxidation state is more stable than +5 ?
 a) Pb b) Al c) Tl d) Bi

More than One correct answer Type Questions

28. Select the correct statement(s)
 a) Sulphur has a lower electron affinity than chlorine
 b) Iodine has a lower electron affinity than bromine
 c) Boron has a lower 1st ionisation energy than phosphorus
 d) Sulphur has a lower 1st ionisation energy than phosphorus
29. Which of the following statement is/are not correct?
 a) I.P. increases down the group
 b) I.P. of s-block element is less than corresponding d-block elements
 c) If Δ I.P. > 16 eV, higher oxidation state is more stable
 d) I.P. of halogen elements is maximum in their respective period
30. How many of the following are endothermic?
 a) $M_{(g)}^- \rightarrow M_{(g)}$ b) $M_{(g)} \rightarrow M_{(g)}^+$ c) $M_{(g)}^+ \rightarrow M_{(g)}^{2+}$ d) $M_{(g)}^{2+} \rightarrow M_{(g)}^{3+}$
31. Which of the following orders for electron affinity is/are correct?
 a) S > O < Se b) Cl > F c) S > O d) O > S
32. Which are correct match?
 a) O < C < S < Se - Atomic size b) Na < Al < Mg < Si - 1st I.P.
 c) MgO < SrO < CS₂O < K₂O - Basic character d) P₄O₁₀ > SO₃ > Cl₂O₇ - Acidic character
33. The incorrect order of decreasing first ionization energy is:
 a) Si > Al > Mg > Na b) Si > Mg > Al > Na
 c) Al > Si > Mg > Na d) Mg > Na > Al > Si

34. Which of the following options are incorrect?
- Decreasing I.E.₂ : F > N > O
 - Increasing atomic size: Be < B < Li < Na
 - Metallic bond strength: Na < Zn < Ca
 - Increasing electron affinity: O²⁻ < O⁻ < O⁺ < O
35. Consider the following ionization steps:
 $X_{(g)} \rightarrow X_{(g)}^+ + e^-; \Delta H = 100 \text{ unit}$; $X_{(g)} \rightarrow X_{(g)}^{2+} + 2e^-; \Delta H = 250 \text{ unit}$
 Select correct statement(s).
- I.E.₁ of X_(g) is 100 unit
 - I.E.₁ of X_(g)⁺ is 150 unit
 - I.E.₂ of X_(g) is 250 unit
 - I.E.₂ of X_(g) is 150 unit
36. Which of the following statements is/are correct
- Van der Waal's radius is always higher than covalent radius
 - All isoelectronic ions belong to the same period
 - IE₁ of N is higher than oxygen while IE₂ of oxygen is higher than nitrogen
 - The electron gain enthalpy of oxygen is higher than that of sulphur.

Linked Comprehension Type Questions

Passage-I :

The minimum amount of energy required to remove the most loosely bound electron from an isolated atom in the gaseous state is known as ionisation energy or ionisation enthalpy (IE₁) of the element. The energy required to remove the second electron from the monovalent cation is called second ionisation enthalpy (IE₂). Similarly, we have third, fourth ionisation enthalpies. The values of ionisation energy depends on a number of factors such as (i) size of the atom (ii) screening effect (iii) nuclear charge (iv) half filled and fully filled orbitals (v) shape of orbital. In a group, the ionisation energy decreases from top to bottom. In a period, the value of ionisation energy increases from left to right.

37. Compared to the second ionisation energy of an atom, the third ionisation energy is
- the same
 - greater
 - smaller
 - half
38. In a period, the ionisation energy is lowest for the :
- noble gases
 - halogens
 - alkaline earth metals
 - alkali metals
39. The first ionisation energy of Mg, Al, P and S follows the order :
- Mg < Al < P < S
 - Al < Mg < P < S
 - Al < Mg < S < P
 - Mg < Al < S < P

Passage-II :

Consider metal (X) and non-metal (Y) react to form ionic compound. Given that metal X is solid at room temperature and non-metal exists as diatomic gaseous molecule.

40. What is the formula of ionic compound if electronic configurations of outermost shell of X = ns² and Y = ns²np³?
- X₃Y₂
 - X₃Y
 - XY₂
 - XY

41. If metal $X_{(s)}$ forms X^+ ion and $Y_{2(g)}$ forms Y^{2-} ion then, which of the following is the correct to calculate enthalpy of following reactions? (if ΔH_f° of X_2Y is exothermic)



- a) $-q = \Delta H_f^\circ(X_2Y) - SE - IP_1 - \frac{BE}{2} + EGE_1 - EGE_2$
 b) $-q = -\Delta H_f^\circ(X_2Y) - 2SE - IP_1 - \frac{BE}{2} - EGE_1 - EGE_2$
 c) $-q = -\Delta H_f^\circ(X_2Y) - 2SE - IP_1 - IP_2 - \frac{BE}{2} + EGE_1 - EGE_2$
 d) $-q = -\Delta H_f^\circ(X_2Y) - 2SE - 2IP_1 - \frac{BE}{2} + EGE_1 - EGE_2$
42. In which of the following cases lattice energy of resulting product will be highest?
- a) X and Y both are monovalent
 b) X and Y both are divalent
 c) X is monovalent and Y is divalent
 d) X is divalent and Y is monovalent

Passage-III :

The reducing effect of the nuclear charge by the inner electrons for an outer electron is termed as shielding (or screening). As a result of shielding, the outer electrons in an atom always experience less nuclear charge than the actual nuclear charge Z . The effective nuclear charge (Z^*) as experienced by an electron is then obtained by subtracting the total shielding contributions from all the other electrons (i.e., except the one under consideration) from the actual nuclear charge.

$$Z^* = Z - s$$

Where s = sum of the shielding contributions. The rules for estimating contributions to s are as follows (Slater's rule)

Contribution to shielding by each electron is :

Electron group	All higher group	Same group	Group $n - 1$	Group $\leq n-2$
1s	0	0.30	–	–
(ns,np)	0	0.35	0.85	1.00
(nd) or (nf)	0	0.35	1.00	1.00

According to Slater's treatment, the energy of an electron in n^{th} shell of an atom having atomic number Z is given by the empirical equation

$$E = -13.6 \left(\frac{Z^*}{n} \right)^2 \text{ eV ; } Z^* = \text{effective nuclear charge}$$

43. Among the following which electron of Fe atom experience minimum attraction from nucleus ? (Atomic number of Fe = 26)
- a) 3d b) 4s c) 2s d) 2p

44. Z^* for a 1s electron in Fe atom is :

- a) 2.85 b) 25.7 c) 25.65 d) 3.75

45. The magnitude of first ionisation energy for Na (according to formula given) is equal to :

- a) energy of its 3s electron
b) energy of its 1s electron
c) energy of its 2s electron
d) energy of its 2p electron

Matrix Matching Type Questions

46. Match the electronic configuration in Column-I with the ionisation energy in Column-II (in KJmole)

Column-I

- A) ns^2
B) ns^2np^1
C) ns^2np^3
D) ns^2np^6

Column-II

- P) 2100
Q) 1400
R) 800
S) 900

47. Column-I

- A) $Li < Na < K < Rb$
B) $Li < Be > B < C$
C) $F_2 > Cl_2 > Br_2 > I_2$
D) $F_2 > O_2 > Cl_2 > S$

Column-II

- P) Increasing order ionisation energy
Q) Decreasing order of non-metallic nature
R) Increasing order of metallic nature
S) Decreasing order of electronegativity

48. Column-I

- A) Exothermic change ($\Delta H = -ve$)
B) Endothermic change ($\Delta H = +ve$)
C) I.E.₂ of oxygen
D) E.A. of oxygen

Column-II

- P) $O_{(g)}^+ \longrightarrow O_{(g)}^{++}$
Q) $O_{(g)} \longrightarrow O_{(g)}^-$
R) $O_{(g)} \longrightarrow O_{(g)}^{++}$
S) $O_{(g)}^- \longrightarrow O_{(g)}^{--}$
T) $O_{(g)} \longrightarrow O_{(g)}^{--}$

Integer Type Questions

49. The first four ionisation energy values of an element are 191, 578, 872, and 8982 K.cal. The number of chlorine atoms per molecule of the compound formed by it is ____

50. Find the sum of the sublimation energy of Li and 1st ionisation energy (in kJ) for Li using the following data. [If your answer is greater than 9 then add all the digits till single digit is obtained]

Bond energy of $F_2 = 154$ kJ/mole

Electron gain enthalpy of F = -328 kJ/mole

Lattice energy of $LiF_{(s)} = 1047$ kJ/mole ΔH_f° for $LiF_{(s)} = -617$ kJ/mole

51. Select total no. of the acidic compounds out of given below
 CsOH , OC(OH)_2 , $\text{SO}_2(\text{OH})_2$, Sr(OH)_2 , Ca(OH)_2 , Ba(OH)_2 , BrOH , NaOH , O_2NOH
52. From the given compounds if X number of compounds are acidic in water.
 CaO , SO_2 , SO_3 , Fe_2O_3 , Cl_2O_7 , CO_2 , Na_2O . Find the value of X.
53. Find the total number of cations for which I.P. of cation is lower than corresponding atom.
 K^+ , O^+ , Ne^+ , P^+ , Be^+ , Na^{2+} , Fe^+ , Sn^{2+}

KEY SHEET (LECTURE SHEET)

EXERCISE-I

LEVEL-I

- 1) 2 2) 4 3) 4 4) 4 5) 3 6) 2 7) 3 8) 3
 9) 1 10) 4 11) 4 12) 3 13) 4 14) 4 15) 2 16) 3
 17) 2 18) 2 19) 1 20) 4 21) 4 22) 4 23) 4 24) 2
 25) 1 26) 2 27) 4 28) 3

LEVEL-II

- 1) b 2) d 3) b 4) b 5) d 6) b 7) d 8) b
 9) b 10) d 11) b 12) b 13) d 14) b 15) d 16) c
 17) d 18) a 19) a 20) c 21) a 22) c 23) d 24) d
 25) acd 26) abc 27) abcd 28) bd 29) ac 30) abcd 31) abc 32) d
 33) b 34) c 35) a 36) b 37) c 38) A-S; B-P; C-R; D-Q
 39) A-Q; B-S; C-T; D-PRS 40) 6 41) 3 42) 1 43) 6 44) 4

EXERCISE-II

LEVEL-I

- 1) 3 2) 1 3) 2 4) 4 5) 4 6) 3 7) 2 8) 1
 9) 4 10) 2 11) 2 12) 2 13) 4 14) 4 15) 2 16) 1
 17) 4 18) 2 19) 4 20) 2 21) 1 22) 3 23) 2 24) 2
 25) 4 26) 1 27) 3 28) 1 29) 4 30) 1 31) 3 32) 3
 33) 2 34) 2 35) 3 36) 1 37) 4 38) 2 39) 1 40) 2

LEVEL-II

- 1) d 2) b 3) c 4) a 5) c 6) b 7) d 8) d
 9) b 10) b 11) a 12) c 13) b 14) b 15) c 16) d
 17) d 18) a 19) d 20) c 21) a 22) d 23) d 24) d
 25) b 26) d 27) d 28) abcd 29) acd 30) abcd 31) abc 32) ab
 33) acd 34) abcd 35) abd 36) ac 37) b 38) d 39) c 40) a
 41) d 42) b 43) b 44) b 45) a 46) A-S; B-R; C-Q; D-P
 47) A-R; B-P; C-Q; D-S 48) A-Q; B-PRST; C-P; D-Q
 49) 3 50) 6 51) 4 52) 4 53) 0

PRACTICE SHEET

EXERCISE-I

(Classification of elements, Mendeleev's periodic table, Modern periodic law, Modern periodic table, Classification of elements into blocks, Classification of elements into types, periodicity, atomic size, atomic radii, ionic radii in groups and periods)

LEVEL-I (MAIN)

Straight Objective Type Questions

- The maximum number of elements available in elemental form
1) 26 2) 92 3) 111 4) 118
- Which of the following is Dobereiner triad
1) Li, Na, K 2) Fe, Co, Ni 3) Ru, Rh, Pd 4) Os, Ir, Pt
- Number of short periods in short form of periodic table
1) 3 2) 2 3) 4 4) 6
- Considering the chemical properties, atomic weight of the element 'Be' was corrected based on
1) Valency 2) Configuration 3) Density 4) Atomic volume
- Eka silicon is now known as
1) Scandium 2) Gallium 3) Germanium 4) Boron
- The atomic weights of "Be" and "In" were corrected by Mendeleef using the formula
1) $\sqrt{v} = a(Z - b)$ 2) $mvr = \frac{nh}{2\pi}$
3) Atomic weight = Equivalent weight x valency 4) Equivalent weight = Atomic weight x valency
- The plot of $\sqrt{v}$ vs Z is
1) Straight line 2) Exponential curve 3) Hyperbolic 4) Curve with -ve slope
- The longest and shortest periods in the periodic table are
1) 1 & 6 2) 2 & 6 3) 6 & 1 4) 1 & 7
- The number of elements present in 2nd, 3rd, 4th and 5th periods of modern periodic table respectively are
1) 2, 8, 8 & 18 2) 8, 8, 18 & 32 3) 8, 8, 18 & 18 4) 8, 18, 18 & 32
- Which of the following pair of elements are from the same group of the periodic table
1) Mg, Cs 2) Mg, Sr 3) Mg, Cl 4) Na, Cl
- Elements of a vertical group have
1) Same atomic number 2) Same electronic configuration
3) Same number of valency electrons 4) Same number of core electrons
- Which of the following pairs of atomic numbers represents elements belonging to the same group?
1) 11, 20 2) 12, 30 3) 13, 31 4) 14, 33

13. As per the modern periodic law, the physical and chemical properties of elements are periodic functions of their
1) atomic number
2) electronic configuration
3) atomic weight
4) atomic size
14. An element with atomic number 20 will be placed in which period of the periodic table?
1) 4
2) 3
3) 2
4) 1
15. If the atomic number of an element is 33, it will be placed in the periodic table in the
1) IA group
2) IIIA group
3) VA group
4) VIIA group
16. The triad not present in Group VIII of Mendeleev's table
1) Li, Na, K
2) Fe, Co, Ni
3) Ru, Rh, Pd
4) Os, Ir, Pt
17. In the periodic table, inversion of atomic weights took place in this pair
1) Argon - Potassium
2) Boron - Scandium
3) Hydrogen - Helium
4) Beryllium - Boron
18. The period that contains only gaseous elements is
1) 1
2) 2
3) 3
4) 4
19. The element which belong to 3rd period and IVA group of periodic table is
1) Silicon
2) Carbon
3) Germanium
4) Tin
20. (A) : According to Mendeleev, periodic properties of elements is a function of their atomic masses.
(R) : Atomic number is equal to number of protons.
1) Both (A) and (R) are true and (R) is the correct explanation of (A)
2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
3) (A) is true but (R) is false
4) (A) is false but (R) is true
21. Pair of elements with the following atomic numbers have the same chemical properties
1) 13 & 22
2) 3 & 11
3) 4 & 24
4) 2 & 1
22. Which of the following is the atomic number of a metal ?
1) 35
2) 34
3) 36
4) 38
23. In which element shielding effect is not possible ?
1) H
2) Be
3) B
4) N
24. Which are correct match ?
i) Eka silicon-Ge ii) Eka aluminium-Ga iii) Eka manganese-Ar iv) Eka scandium-B
1) ii, iii
2) i, ii, iii
3) i, iv
4) i, ii
25. Which of the following is true about the element $_{33}\text{As}$?
1) It is the 5th period element
2) It is p-block element
3) It belongs to 16th group
4) It is the member of VIA group
26. What would be the atomic number of the next halogen if discovered in future?
1) 115
2) 119
3) 117
4) 121
27. The characteristic properties of transition elements are due to
1) Unpaired electrons in d-subshell
2) d-orbitals have five fold degeneracy
3) Presence of 2 nodal planes for d-orbital
4) Because they belong to d-block

28. Rare earths are generally
 1) Actinides 2) f-Block elements
 3) Inner transition elements 4) Lanthanides
29. Lanthanum element with $z = 57$ belongs to
 1) s-block 2) p-block 3) d-block 4) f-block
30. In the periodic table transition elements begin with
 1) Scandium 2) Zinc 3) Copper 4) Mercury
31. Inert gas element which has a different valence shell configuration
 1) Xe 2) Ne 3) Kr 4) He
32. Atomic numbers of actinides are
 1) 57 to 71 2) 80 to 103 3) 58 to 71 4) 90 to 103
33. Most of the non-metals are present in the long form of the periodic table in
 1) p-block 2) f-block 3) d-block 4) s-block
34. Metal used as catalyst in the hydrogenation of vegetable oils
 1) Iron 2) Molybdenum 3) Nickel 4) Sodium
35. The 4f-subshell is successively filled for
 1) Rare earths 2) Rare gases 3) Transition metals 4) Alkaline earth metals
36. The role of 'Molybdenum' in Haber's synthesis is
 1) A positive catalyst 2) A negative catalyst 3) Poison for catalyst 4) Promoter for catalyst
37. The period in which s-block, p-block and d-block elements are present
 1) 1 2) 6 3) 7 4) 3
38. Which of the following is not a periodic property
 1) Valency 2) Specific heat 3) Ionisation potential 4) Atomic size
39. Atomic radius is measured by
 1) Mulliken oil drop method 2) Rutherford's α -ray scattering experiment
 3) X-ray diffraction technique 4) Electric discharge tube experiment
40. Atomic radius depends upon
 i) Number of bonds formed by the atom ii) Nature of the bonding
 iii) Oxidation state of the atom
 1) i and ii 2) ii and iii 3) i and iii 4) i, ii, iii
41. Largest ion among the following is
 1) Na^+ 2) O^{2-} 3) S^{2-} 4) Cl^-
42. Which of the following has the largest atomic radius ?
 1) Al 2) Si 3) Cl 4) Na
43. Which of the following order is correct for the size of Fe^{3+} , Fe and Fe^{2+} ?
 1) $\text{Fe} < \text{Fe}^{2+} < \text{Fe}^{3+}$ 2) $\text{Fe}^{2+} < \text{Fe}^{3+} < \text{Fe}$
 3) $\text{Fe} < \text{Fe}^{3+} < \text{Fe}^{2+}$ 4) $\text{Fe}^{3+} < \text{Fe}^{2+} < \text{Fe}$

44. Metallic radius of Ca is 200pm. Covalent radius of Ca is
 1) 200 pm 2) 230 pm 3) 280 pm d) 174 pm
45. Similarity in the radius of Zr and Hf is explained on the basis of
 1) Lanthanide contraction 2) Inert pair effect
 3) Same outershell configuration 4) Anomalous configuration
46. The correct sequence which shows decreasing order of the ionic radii of the elements is
 1) $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+} > \text{O}^{2-} > \text{F}^-$ 2) $\text{Na}^+ > \text{F}^- > \text{Mg}^{2+} > \text{O}^{2-} > \text{Al}^{3+}$
 3) $\text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$ 4) $\text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+ > \text{F}^- > \text{O}^{2-}$
47. The correct order of radii is :
 1) $\text{N} < \text{Be} < \text{B}$ 2) $\text{Mg}^{2+} < \text{Li}^+ < \text{N}^{3-}$ 3) $\text{Na} < \text{Li} < \text{K}$ 4) $\text{Fe}^{+3} < \text{Fe}^{2+} < \text{Fe}^{4+}$
48. K^+ , Cl^- , Ca^{2+} , S^{2-} ions are isoelectronic. The decreasing order of their size is :
 1) $\text{S}^{2-} > \text{Cl}^- > \text{K}^+ > \text{Ca}^{2+}$ 2) $\text{Ca}^{2+} > \text{K}^+ > \text{Cl}^- > \text{S}^{2-}$ 3) $\text{K}^+ > \text{Cl}^- > \text{Ca}^{2+} > \text{S}^{2-}$ 4) $\text{Cl}^- > \text{S}^{2-} > \text{Ca}^{2+} > \text{K}^+$
49. Select correct statement :
 1) Across a transition series (from Cr to Cu), there is only a small decrease in atomic radius from one element to another due to very small increase in effective nuclear charge
 2) The rate of decrease in the size across the lanthanide series is less than the across the first transition series
 3) Both are correct statements 4) None of the statement is correct
50. The lanthanides contraction refers to :
 1) ionic radius of the series 2) valence electrons of the series
 3) the density of the series 4) electronegativity of the series
51. Which of the following statement is correct ?
 1) Metallic radius refer to metals only and is greater than covalent radius
 2) Metallic radius refer to metals only and is smaller than covalent radius
 3) Generally covalent radius refer to non-metals as well as metals in bonded state (covalent bond).
 4) Atomic radii of noble gases are expressed as van der Waal's radii which are smaller than metallic radii.
52. Which one among the following sets of ions represents the collection of isoelectronic species ?
 1) S^{2-} , Cl^- , K^+ , Ca^{2+} , Sc^{3+} 2) N^{3-} , O^{2-} , Na^+ , Mg^{2+} , Al^{3+}
 3) K^+ , Cl^- , Mg^{2+} , Al^{3+} , Sc^{3+} 4) 1 and 2 both
53. If each orbital can take maximum of three electrons, the number of elements in the third period of the periodic table will be :
 1) 12 2) 6 3) 8 4) 24

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. The sub-shells filled one by one for 4th period elements are
 a) 3d, 4s and 4p b) 4s, 4p and 4d c) 4s, 3d and 4p d) 3d, 4p and 4s
2. The starting element and last element in the largest period in modern periodic table are
 a) Rb and Xe b) Cs and I c) Cs and Rn d) Fr and Kr

3. Which of the following has both members from the same period of the periodic table
 a) Na, F b) Mg, Ca c) Na, Cl d) Be, Al
4. Atomic number of nitrogen is 7. The atomic number of the third member in the same family is
 a) 23 b) 15 c) 33 d) 51
5. Element with atomic number 38, belongs to
 a) II A group and 5th period b) II A group and 2nd period
 c) V A group and 2nd period d) III A group and 5th period
6. Set of elements with the following atomic numbers belong to the same group
 a) 9, 16, 35, 3 b) 12, 20, 4, 38
 c) 11, 19, 27, 5 d) 24, 47, 42, 55
7. Which of the following pairs has both members from the same group of the periodic table?
 a) Mg-Ba b) Mg-Na c) Mg-Cu d) Mg-Cl
8. The elements with atomic number 10, 18, 36, 54 and 86 are all
 a) Light metals b) Inert gases c) Halogens d) Rare earths
9. Which of the following pairs has elements containing same number of electrons in the outermost orbit?
 a) N, O b) Na, Cl c) Ca, Cl d) Cl, Br
10. The period that includes all blocks of elements is
 a) 1 b) 2 c) 6 d) 7
11. Among s-block metals and transition metals, which are more metallic?
 a) s-block metals b) Transition metals
 c) Both are equally metallic d) Cannot be predicted
12. Element with atomic number 52 belongs to
 a) s-block b) p-block c) d-block d) f-block
13. The general electronic configuration of f-block elements is
 a) $ns^2 np^6 (n-1)d^{0-1} (n-2)f^{1-14}$ b) $ns^2 (n-1)d^{0-1} (n-2)f^{1-14}$
 c) $ns^2 nd^{0-1} nf^{1-14}$ d) $ns^2 (n-1)d^{0-1} (n-1)f^{1-14}$
14. What is the position of the elements in the periodic table satisfying the electronic configuration $(n-1)d^1 ns^2$ for $n = 4$?
 a) 3rd period and 3rd group b) 4th period and 3rd group
 c) 3rd period and 2nd group d) 4th period and 2nd group
15. Elements A, B, C, D and E have the following electronic configurations:
 A) $1s^2, 2s^2 2p^1$ B) $1s^2, 2s^2 2p^6, 3s^2, 3p^1$
 C) $1s^2, 2s^2 2p^6, 3s^2, 3p^3$ D) $1s^2, 2s^2 2p^6, 3s^2, 3p^5$
 E) $1s^2, 2s^2 2p^6, 3s^2, 3p^6$
 Which among these will belong to same group in the periodic table?
 a) (A) and (C) b) (A) and (D) c) (A) and (B) d) (A) and (E)

16. The outer electronic configuration of an element is, $(n-2)f^7(n-1)d^1ns^2$. To which block in the long form of the periodic table does this belong?
 a) s-block b) p-block c) d-block d) f-block
17. The elements with atomic numbers 90 to 103 are known as :
 a) d-block elements b) lanthanides
 c) actinides d) transition elements
18. In Mendeleev's periodic table, silver belongs to IB group. The group to which silver belongs in long form of periodic table is :
 a) first b) eleventh c) tenth d) sixteenth
19. Match the following lists and select the correct answer.
- | List-I | | | | List-II | | | |
|--|------------|------------|------------|--------------------|--|--|--|
| A) $1s^2, 2s^2 2p^6, 3s^2 3p^6, 4s^1$ | | | | 1) d-block element | | | |
| B) $1s^2, 2s^2 2p^6, 3s^2 3p^6$ | | | | 2) Halogen | | | |
| C) $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^6, 4s^2$ | | | | 3) Alkali metal | | | |
| D) $1s^2, 2s^2 2p^5$ | | | | 4) Noble gas | | | |
| A B C D | A B C D | A B C D | A B C D | | | | |
| a) 1 2 3 4 | b) 3 4 1 2 | c) 1 3 2 4 | d) 2 4 3 1 | | | | |
20. Atomic radii of fluorine and neon in angstrom units are respectively
 a) 0.72, 1.62 b) 0.72, 0.72 c) 1.2, 1.2 d) 1.62, 0.72
21. Which of the following will have largest size?
 a) Br b) I^- c) I d) F
22. The size of the following species increases in the order
 a) $Mg^{2+} < Na^+ < F^- < Al^{3+}$ b) $Al^{3+} < Mg^{2+} < Na^+ < F^-$
 c) $Na^+ < F^- < Al^{3+} < Mg^{2+}$ d) $Na^+ < Al^{3+} < Mg^{2+} < F^-$
23. In a period of representative elements, the decrease in ionic radius when compared with the corresponding decrease in atomic radius
 a) is equal b) is less
 c) is more d) Cannot be predicted
24. In which of the following sets, elements have nearly same atomic radii?
 a) Li, Be, B b) Mg, Ca, Sr c) Fe, Co, Ni d) O, S, Se
25. Correct order of atomic radii is
 a) $N < C < P < S$ b) $C < N < S < P$
 c) $C < N < P < S$ d) $N < C < S < P$
26. If an element 'X' is assumed to have the types of radii, then their order is
 a) Crystal radius > Van der waals radius > Covalent radius
 b) van der waals radius > Crystal radius > Covalent radius
 c) Covalent radius > Crystal radius > van der waals radius
 d) Van der waals radius > Covalent radius > Crystal radius

27. Which of the following has smallest radius?
 a) Be^{+2} b) Li^+ c) O^{-2} d) F^-
28. In general, the configuration of lanthanides is $(n-2)f^{1-14}(n-1)s^2p^6d^{0-1}ns^2$. It has been observed that, with increase in atomic number of lanthanides, there is a gradual decrease in ionic radii from $\text{La}(1.22\text{\AA})$ to $\text{Lu}(0.99\text{\AA})$. The reason for decrease in ionic radii is an increase in
 a) Electronegative character
 b) Valency electrons and number of shells
 c) Atomic and ionic volumes
 d) Nuclear attraction for valence electrons leading to inward shrinking of shell
29. An increase in both atomic and ionic radii with atomic number occurs in any group of the periodic table and in accordance with this the ionic radii of Ti(IV) and Zr(IV) ions are $0.68\text{\AA}$ and $0.74\text{\AA}$ respectively; but for Hf(IV) ion, the ionic radius is $0.75\text{\AA}$, which is almost the same as that for Zr(IV) ion. This is due to
 a) Greater degree of covalency in compounds of Hf^{4+}
 b) Lanthanide contraction
 c) Actinide contraction
 d) Difference in co-ordination number of Zn^{4+} and Hf^{4+} in their compounds
30. Mercury is the only metal which is liquid at 0°C . This is due to its
 a) Very high ionisation energy and weak metallic bond
 b) Low ionisation potential and high electronegativity
 c) High atomic mass and small size
 d) High electronegativity and low ionisation potential
31. What is the position of the element in the periodic table satisfying the electronic configuration $(n-1)d^1 ns^2$ for $n=4$.
 a) 3rd period and 3rd group b) 4th period and 4th group
 c) 3rd period and 2nd group d) 4th period and 3rd group
32. In Mendeleev's periodic table, silver belongs to IB group. The group to which silver belongs in long form of periodic table is (Atomic number = 47) :
 a) first b) tenth c) sixteenth d) eleventh
33. The correct decreasing order of atomic size among the following species is: $\text{Ar}, \text{K}^+, \text{Cl}^-, \text{S}^{2-}, \text{Ca}^{2+}$
 a) $\text{Ca}^{2+} > \text{K}^+ > \text{Ar} > \text{Cl}^- > \text{S}^{2-}$ b) $\text{K}^+ > \text{Ca}^{2+} > \text{Cl}^- > \text{Ar} > \text{S}^{2-}$
 c) $\text{S}^{2-} > \text{Cl}^- > \text{Ar} > \text{K}^+ > \text{Ca}^{2+}$ d) $\text{S}^{2-} > \text{Ar} > \text{Cl}^- > \text{Ca}^{2+} > \text{K}^+$
34. Arrange $\text{Ce}^{3+}, \text{La}^{3+}, \text{Pm}^{3+}$ and Yb^{3+} in increasing order of their size
 a) $\text{Yb}^{3+} < \text{Pm}^{3+} < \text{Ce}^{3+} < \text{La}^{3+}$ b) $\text{Ce}^{3+} < \text{Yb}^{3+} < \text{Pm}^{3+} < \text{La}^{3+}$
 c) $\text{Yb}^{3+} < \text{Pm}^{3+} < \text{La}^{3+} < \text{Ce}^{3+}$ d) $\text{Pm}^{3+} < \text{La}^{3+} < \text{Ce}^{3+} < \text{Yb}^{3+}$
35. The atomic radii of transition elements from Cr to Cu are almost equal because
 a) Increased effective nuclear charge is balanced by decreased screening effect of $(n-1)d$ orbitals
 b) Increased effective nuclear charge is balanced by increased screening effect of $(n-1)d$ orbitals
 c) Decreased effective nuclear charge is balanced by increased screening effect of $(n-1)d$ orbitals
 d) Decreased effective nuclear charge is balanced by decreased screening effect of $(n-1)d$ orbitals

36. Astatine is a radioactive halogen. It is a solid at room temperature because
- a) of greater van der Waal's force of attraction between large atoms of astatine
 - b) of less van der Waal's force of attraction between large atoms of astatine
 - c) of less van der Waal's force of attraction between small atoms of astatine
 - d) it shows non-metallic characters
37. The statement that is false for the long form of the periodic table is
- a) It reflects the sequence of filling the electrons in the order of sub energy levels s,p,d and f
 - b) It helps to predict the stable valency states of the element
 - c) It reflects trends in physical and chemical properties of the elements
 - d) It helps to predict the relative ionicity of the bond between any two elements
38. The transition elements (d-block elements) show variable oxidation states because
- a) of the presence of ns, np and nd electrons
 - b) the energy difference between (n-1)d and ns electrons is very less, thus (n-1)d electrons also behave like valence electrons.
 - c) of the presence of ns and nd orbitals
 - d) of the presence of electrons in np and nd orbitals

More than One correct answer Type Questions

39. Which of the following outer electronic configurations corresponds to the d-block of the periodic table ?
- a) $3s^2 3p^3$
 - b) $3d^5 4s^1$
 - c) $3s^2 3p^6 4s^2$
 - d) $5d^1 6s^2$
40. Which of the following species have five unpaired electrons ?
- a) Cr^{3+}
 - b) Mn^{2+}
 - c) Fe^{3+}
 - d) Cu^{2+}
41. The first element of a group in many ways differ from the other heavier members of the group. This is due to
- a) the small size
 - b) the high electronegativity
 - c) high ionisation potential
 - d) the unavailability of d-orbitals
42. Chlorine atom does not differ from chloride ion in the number of which of the following?
- a) Protons
 - b) Neutrons
 - c) Electrons
 - d) Size
43. In which of the following pairs the radius of the second species is smaller than that of the first ?
- a) Na, Mg
 - b) O^{2-} , N^{3-}
 - c) Li^+ , Be^{2+}
 - d) Ba^{2+} , Sr^{2+}
44. Which of the following is the correct statements
- a) All the actinide elements are radioactive
 - b) Alkali and alkaline earth metals are s-block elements
 - c) Chalcogens and halogens are p-block elements
 - d) The first member of the lanthanide series is lanthanum.
45. Which of the following statements related to the modern periodic table is/are correct
- a) The p-block has 6 vertical columns, i.e., groups
 - b) The d-block has 8 vertical columns
 - c) Each block contains a number of columns equal to the number of electrons that can occupy that subshell
 - d) The block indicates value of azimuthal quantum number (*l*) for the last subshell that received electrons in building up the electronic configuration.

46. Stability of ions of Ge, Sn and Pb will be in the order
 a) $\text{Ge}^{2+} < \text{Sn}^{2+} < \text{Pb}^{2+}$ b) $\text{Ge}^{4+} > \text{Sn}^{4+} > \text{Pb}^{4+}$ c) $\text{Sn}^{4+} > \text{Sn}^{2+}$ d) $\text{Pb}^{2+} > \text{Pb}^{4+}$
47. Which of the following is/are correct for Mendeleev's periodic table
 a) There are nine groups including zero group
 b) All the groups are divided into two subgroups
 c) The group number indicates the valency of the elements on oxygen scale
 d) The group number indicates the oxidation number of the elements
48. Which of the following statement(s) is(are) correct?
 a) vander Waals radius of iodine is more than its covalent radius
 b) All isoelectronic ions belong to the same period of the periodic table
 c) IE_1 of N is higher than that of O while IE_2 of O is higher than that of N
 d) The electron gain enthalpy of N is positive while that of P is negative.
49. Which of the following is/are the effect of f-orbital contraction effect?
 a) Similarity in the size of Zr, Hf
 b) High electron affinities among the post lanthanides
 c) High I. E. of Hg as compare of Tl
 d) High I. E. of Tl as compare to In
50. Which is incorrect order of size of O , O^{2-} , F^- and F ?
 a) $\text{O}^{2-} > \text{O} > \text{F}^- > \text{F}$ b) $\text{O} > \text{O}^{2-} > \text{F} > \text{F}^-$ c) $\text{O}^{2-} > \text{F}^- > \text{F} > \text{O}$ d) $\text{O}^{2-} > \text{F}^- > \text{O} > \text{F}$

Linked Comprehension Type Questions

Passage-I :

In the modern periodic table, the elements are placed in order to increasing atomic number. These have been numerous designs of the table over the years but the most common is the long form of periodic table. The long form of periodic table shows all the elements in numerical order

51. Write the expected ground state electronic configuration of the element with one unpaired 5p electron that forms a covalent compound with fluorine.
 a) $[\text{Kr}]5s^2, 5p^5$ b) $[\text{Kr}]5s^2, 4d^{10} 5p^1$ c) $[\text{Kr}]5s^2, 4d^{10}, 5p^5$ d) $[\text{Ar}]5s^2 4d^{10} 5p^5$
52. What is the atomic number of the (as yet undiscovered) alkali earth metal after radium?
 a) 120 b) 121 c) 124 d) 118
53. Which of the following inert gas in which electrons are occupying in 4f- orbitals but no electron in 6d- orbitals in ground state electronic configuration
 a) Kr b) Xe c) Rn d) Uuo

Passage-II :

It is not possible to measure the atomic radius precisely since the electron cloud surrounding the atom does not have a sharp boundary. One practical approach to estimate the size of an atom of a non-metallic element is to measure the distance between two atoms when they are bound together by a single bond in a covalent molecule and then dividing by two. For metals we define the term "metallic radius" which is taken as half the internuclear distance separating the metal cores in the metallic crystal. The van der waal's radius represents the over all size of the atoms which includes its valence shell in a non bonded situation.

54. The size of isoelectronic species - F^- , Na^+ and Mg^{2+} is effected by :
- nuclear charge
 - valence principal quantum number (n)
 - electron-electron interaction in the outer orbitals
 - none of these
55. The correct order of radii is :
- $Na < Li < K$
 - $O > S > Se$
 - $Cl < F < Li$
 - $Fe^{3+} < Fe^{2+} < Fe$
56. Atomic radii of the noble gases are larger than the precedent elements of the same periods because :
- atomic radius of a noble gas is expressed as vander Waal's radius.
 - valence shell electrons are completely filled so there is interelectronic repulsions
 - both (a) and (b)
 - none

Matrix Matching Type Questions

57. Column-I

- $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
- $1s^2 2s^2 2p^6 3s^2 3p^6$
- $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$
- $1s^2 2s^2 2p^5$

Column-II

- d-block element
- Halogen
- Alkali metal
- Noble gas

58. Column-I

- 7th period
- IIIB group
- IA
- VIIA

Column-II

- Transition elements
- Inner transition elements
- Hydrogen
- Halogens

59. Column-I

- Sulphur atom
- Bromine atom
- Rubidium atom
- Iron atom

Column-II

- Presence of unpaired electron(s) in p-orbital(s)
- All electron in s-orbital are paired
- MF_2 type compound formation (F = Fluorine) (M = given atom)
- Most electropositive element out of given elements
- Magnetic moment $> \sqrt{3}$ B.M.

Integer Type Questions

- All the four blocks and four types of elements are present in the ____ period
- The maximum number of valency electrons possible for atoms in the second period of periodic table is ____
- Maximum number of elements are present in IIB group. Among them number of d-block elements is (are) ____
- Liquid non-metallic element is present in ____ period of long form of periodic table
- The number of groups which contains only gaseous elements is ____
- The number of elements present in each short periods are ____
- How many elements are possible for 1st period of periodic table if azimuthal quantum number can have integral values from 0, 1, 2... (n + 1)?

EXERCISE-II

(Ionisation potential, Factors affecting Ionisation potential, Ionisation enthalpy in groups and periods, Electron gain enthalpy (electron affinity) in groups and periods, Electro negativity, EN scales, EN values in groups and periods, Valency and oxidation states, Electro positivity, metallic properties, Acidic and basic strength of oxides and oxy acids and hydric acids, Chemical reactivity of elements and diagonal relationship)

LEVEL-I (MAIN)

Straight Objective Type Questions

- The element that possess the lowest ionisation energy among the following
 1) Oxygen 2) Fluorine 3) Sulphur 4) Nitrogen
- Atoms of the following group possess the highest ionisation energies
 1) IA 2) IIA 3) VA 4) Zero
- Atoms of the following group possess the lowest ionisation energies
 1) IA 2) IIA 3) VA 4) Zero
- Configuration of the element with the highest ionisation energy is
 1) $[\text{Ne}]3s1$ 2) $[\text{Ne}] 3s^2 3p^3$ 3) $[\text{Ne}] 3d^{10} 4s^2 4p^3$ 4) $[\text{Ne}] 3s^2 3p^4$
- When Lithium vapour is taken in a discharge tube and the potential difference between the electrodes is 5.4 eV, there is a sudden increase in the flow of current. The ionisation energy of Lithium is
 1) 54 eV 2) 520 kJ mol⁻¹ 3) 54 kJ atom⁻¹ 4) 5.4 eV atom⁻¹
- The five successive ionisation energies of the element are 800, 2427, 3658, 25024 and 32824 kJmol⁻¹ respectively. The number of valence electrons is
 1) 3 2) 5 3) 4 4) 2
- Which has highest second ionisation energy
 1) N 2) C 3) O 4) F
- Which of the following element has the greatest tendency to lose electrons ?
 1) F 2) S 3) Fe 4) Be
- Second ionisation potential of oxygen is
 1) Equal to that of fluorine 3) Less than that of fluorine
 2) Greater than that of fluorine 4) Half of that of fluorine
- Among the elements A, B, C and D having atomic numbers 9, 10, 11 and 12 respectively the correct order of ionisation energies is
 1) $A > B > C > D$ 2) $B > A > D > C$ 3) $B > A > C > D$ 4) $D > C > B > A$
- Configuration that shows the highest energy released when an electron is added to the atom is
 1) $1s^2 2s^2 2p^3$ 2) $1s^2 2s^2 2p^4$ 3) $1s^2 2s^2 2p^5$ 4) $1s^2 2s^2 2p^6$
- Electron affinity of Fluorine is less than that of Chlorine because
 1) Electronegativity of Fluorine is more 2) 2p sub shell of F is smaller
 3) Chlorine is a stronger oxidant 4) Bond dissociation energy of F₂ is less

13. Among chalcogens electron affinity is highest for
1) O 2) S 3) Se 4) Te
14. The element with highest electron affinity is
1) Fluorine 2) Cesium 3) Helium 4) Chlorine
15. Incorrect statement is
1) Fluorine has the highest electron affinity
2) Greater the nuclear charge, greater is the electron affinity
3) The electron affinity of Nitrogen is positive (energy is absorbed)
4) Chlorine has highest electron affinity
16. The correct order of electron affinity of the elements of oxygen family in the periodic table is
1) $O > S > Se$ 2) $S > O > Se$ 3) $S > Se > O$ 4) $Se > O > S$
17. Which of the following is the correct order of electron affinity
1) $I > Br > F > Cl$ 2) $F < Cl < Br < I$ 3) $F > Cl > Br > I$ 4) $I < Br < F < Cl$
18. The electronegativity values according to Mulliken scale are ___ times to those in Pauling scale
1) 0.208 2) 2 3) 2.8 4) 544
19. Electronegativity on Mulliken scale is limited to
1) Monovalent atoms 2) Bivalent atoms
3) Both monovalent and bivalent atoms 4) All multivalent atoms
20. If I and E are ionisation energy and electron affinity of an element in kJ mole^{-1} electronegativity is given as
1) $\frac{I+E}{2}$ 2) $\frac{I+E}{5.6}$ 3) $\frac{I+E}{129}$ 4) $\frac{I+E}{544}$
21. In a period electronegativity is highest for
1) Chalcogen 2) Halogen 3) Inert gas 4) Alkali metal
22. The values that are useful in writing chemical formulae and in calculation of oxidation states are
1) Ionisation potential 2) Electron affinity 3) Electronegativity 4) Metallic character
23. Elements with high electronegativity are generally
1) Good reductants 2) Hard solids 3) Good oxidants 4) Soft solids
24. The formula of a metallic carbonate is MCO_3 . The formula of that metallic perchlorate is
1) MClO_4 2) M_2ClO_4 3) M_3ClO_4 4) $\text{M}(\text{ClO}_4)_2$
25. A metal forms a chloride with the formula MCl_2 . Formula of Phosphoric acid is H_3PO_4 . Formula of the Phosphate of the metal is
1) M_3PO_4 2) MPO_4 3) $\text{M}_3(\text{PO}_4)_2$ 4) M_2PO_4
26. Among the following elements most acidic oxide is given by
1) Al 2) P 3) N 4) Sb
27. The strongest reducing agent is
1) K 2) Al 3) Mg 4) Br

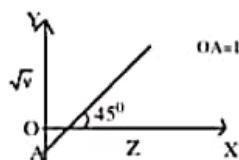
28. The more basic oxide is
 1) CaO 2) MgO 3) K_2O 4) Na_2O
29. An element with $1s^2 2s^2 2p^6 3s^2$ electronic arrangement will be forming
 1) Acidic oxide 2) Basic oxide 3) Neutral oxide 4) Amphoteric oxide
30. (A) : Li generally forms covalent compounds.
 (R) : Li^+ ion is small and has high polarizing power.
 1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 3) (A) is true but (R) is false 4) (A) is false but (R) is true
31. (A) : As_2O_3 and Sb_2O_3 are amphoteric in nature.
 (R) : As and Sb are metalliods.
 1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 3) (A) is true but (R) is false 4) (A) is false but (R) is true
32. (A) : Lithium resembles magnesium in its properties.
 (R) : The ratio of ionic charge to (Ionic radius)² is almost same for Li^+ and Mg^{2+} .
 1) Both (A) and (R) are true and (R) is the correct explanation of (A)
 2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 3) (A) is true but (R) is false 4) (A) is false but (R) is true
33. Some statements are given. Among them the correct statements are
 i) IP_2 of sodium is greater than that of Magnesium
 ii) IP_2 of lithium is greater than IP_1 of Helium
 iii) IP_2 of sodium is greater than IP_1 of Neon
 iv) IP_1 of oxygen is greater than that of Nitrogen
 1) All are correct 2) Only i, ii and iii are correct
 3) Only i, ii are correct 4) Only i, iv are correct
34. Elements X, Y and Z have atomic numbers 19, 37 and 55 respectively. Which of the following statements is true about them?
 1) Their ionization potential would increase with increasing atomic number
 2) 'Y' would have an ionization potential between those of X and Z.
 3) Z would have the highest ionization potential
 4) Y would have the highest ionization potential

LEVEL-II (ADVANCED)

Straight Objective Type Questions

1. The correct decreasing order of electropositive character among the following elements is: Fe, Sc, Rb, Br, Te, F, Ca
 a) $Fe > Sc > Rb > Br > Te > F > Ca$ b) $Ca > Rb > Sc > Fe > Te > F > Br$
 c) $Rb > Ca > Sc > Fe > Br > Te > F$ d) $Rb > Ca > Sc > Fe > Te > Br > F$

2. Which of the following has the highest electron gain enthalpy
 (A) $[\text{Ne}]3s^23p^3$ (B) $[\text{Ne}]3s^23p^4$ (C) $[\text{Ne}]3s^23p^5$ (D) $[\text{Ne}]3s^23p^2$
 a) A b) B c) C d) D
3. The compound of vanadium has magnetic moment of 1.73 BM. The vanadium chloride has the formula:
 a) VCl_2 b) VCl_3 c) VCl_4 d) VCl_5
4. The frequency of the characteristic X ray of K_α line of metal target 'M' is 2500 cm^{-1} and the graph between $\sqrt{\nu}$ vs 'Z' is as follows, then atomic number of M is



- a) 49 b) 50 c) 51 d) 25
5. $(\text{IE})_1$ and $(\text{IE})_2$ of $\text{Mg}_{(g)}$ are $740, 1450 \text{ kJ mol}^{-1}$. Calculate percentage of $\text{Mg}^+_{(g)}$ and $\text{Mg}^{2+}_{(g)}$ if 1 g of $\text{Mg}_{(g)}$ absorbs 50.0 kJ of energy.
 a) $\% \text{Mg}^+ = 70\%$ $\% \text{Mg}^{+2} = 30\%$ b) $\% \text{Mg}^+ = 68.35\%$ $\% \text{Mg}^{+2} = 31.65\%$
 c) $\% \text{Mg}^+ = 72\%$ $\% \text{Mg}^{+2} = 28\%$ d) $\% \text{Mg}^+ = 60\%$ $\% \text{Mg}^{+2} = 40\%$
6. Use (IE) and (EA) listed below to determine whether the following process is endothermic exothermic.
 $\text{Mg}_{(g)} + 2\text{F}_{(g)} \rightarrow \text{Mg}^{2+}_{(g)} + 2\text{F}^{-}_{(g)}$
 $(\text{IE})_1$ of $\text{Mg}_{(g)} = 737.7 \text{ kJ mol}^{-1}$; $(\text{IE})_2$ of $\text{Mg}_{(g)} = 1451 \text{ kJ mol}^{-1}$; (EA) of $\text{F}_{(g)} = -328 \text{ kJ mol}^{-1}$
 a) Exothermic b) Endothermic c) Both d) can't be predicted
7. If Aufbau rule is not followed, K ($Z=19$) will be placed in
 a) s-Block b) p-Block c) d-Block d) f-Block
8. $\text{M}_{(g)} \rightarrow \text{M}^+_{(g)} + e^-$, $\Delta H = 100 \text{ eV}$; $\text{M}_{(g)} \rightarrow \text{M}^{2+}_{(g)} + 2e^-$, $\Delta H = 250 \text{ eV}$
 Which is incorrect statement?
 a) I_1 of $\text{M}_{(g)}$ is 100 eV b) I_1 of $\text{M}^+_{(g)}$ is 150 eV
 c) I_2 of $\text{M}_{(g)}$ is 250 eV d) I_2 of $\text{M}_{(g)}$ is 150 eV
9. AB is predominantly ionic as $A^+ B^-$ if
 a) $(\text{IP})_A < (\text{IP})_B$ b) $(\text{EA})_A < (\text{EA})_B$ c) $(\text{EN})_A < (\text{EN})_B$ d) $(\text{IP})_B < (\text{IP})_A$
10. EN of the element (A) is E_1 and IP is E_2 . Hence EA will be
 a) $2E_1 - E_2$ b) $E_1 - E_2$ c) $E_1 - 2E_2$ d) $(E_1 + E_2)/2$
11. The amount of energy released when 10^6 atoms of iodine in vapour state are converted to I^+ ions is $4.9 \times 10^{-13} \text{ J}$. What is the electron affinity of iodine in eV per atom
 a) 2.0 b) 2.5 c) 2.75 d) 3.06
12. Gd (64) has ___ unpaired electrons with sum of spin ___
 a) 7, 3.5 b) 8, 3 c) 6, 3 d) 8, 4
13. For the process, $\text{X}_{(g)} + e^- \rightarrow \text{X}^{-}_{(g)}$, $\Delta H = x$ and $\text{X}^{-}_{(g)} \rightarrow \text{X}_{(g)} + e^-$, $\Delta H = y$ Select correct alternate:
 a) ionisation energy of $\text{X}^{-}_{(g)}$ is y b) electron affinity of $\text{X}_{(g)}$ is x
 c) electron affinity of $\text{X}_{(g)}$ is $-y$ d) all are correct statements.

14. Some statements are given regarding nature of oxides
- In second period, nitrogen form strongest acidic oxide
 - In third period, sodium forms strongest basic oxide
 - Oxides of metalloids are generally amphoteric in nature
- i and iii are correct
 - ii and iii are correct
 - i and ii are correct
 - i, ii and iii are correct
15. (A), (B), and (C) are elements in the third short period. Oxide of (A) is ionic, that of (B) is amphoteric and of (C) a gaint molecule. (A), (B) and (C) have atomic number in the order
- (A) < (B) < (C)
 - (C) < (B) < (A)
 - (A) < (C) < (B)
 - (B) < (A) < (C)
16. Match the List-I with List-II and select the correct option
- | List-I | | | | List-II | | | |
|--|--|--|--|---------------------|--|--|--|
| A) Most electronegative element | | | | 1) Chlorine | | | |
| B) The element having highest electron gain enthalpy | | | | 2) Hydrogen | | | |
| C) Most abundant element in the universe | | | | 3) Nitrogen | | | |
| D) Most abundant gas in atmosphere | | | | 4) Fluorine | | | |
| A B C D | | | | A B C D | | | |
| a) 1 2 3 4 | | | | b) 4 3 2 1 | | | |
| | | | | c) 4 1 2 3 | | | |
| | | | | d) 2 3 4 1 | | | |
17. Beryllium shows diagonal relationship with aluminium. Which of the following similarity is incorrect?
- Be_2C like Al_4C_3 yields methane on hydrolysis
 - Be, like Al is rendered passive by HNO_3
 - $\text{Be}(\text{OH})_2$ like $\text{Al}(\text{OH})_3$ is basic
 - Be forms beryllates and Al forms aluminate
18. Diagonal relationship is shown by
- B - S
 - Li - Mg
 - Mg - Ca
 - S - Se
19. Aqueous solutions of two compounds $\text{M}_1 - \text{O} - \text{H}$ and $\text{M}_2 - \text{O} - \text{H}$ are prepared in two different beakers. If the electronegativity of $\text{M}_1 = 3.4$, $\text{M}_2 = 1.2$, $\text{O} = 3.5$ and $\text{H} = 2.1$, then the nature of the two solutions will be respectively.
- Acidic, basic
 - Acidic, acidic
 - Basic, acidic
 - Basic, basic
20. Nitrogen and phosphorous belong to the same group in the periodic table and yet the later forms H_3PO_3 , H_3PO_4 , $(\text{HPO}_3)_n$, $\text{H}_4\text{P}_2\text{O}_7$ and the former forms only HNO_3 , HNO_2 and HNO_4 i.e. former forms less number of oxyacids. This is because
- N is much more electronegative than P
 - N atom is smaller in size compared to P-atom
 - N atom does not have d-orbitals in its valence shell but P has
 - N has a lower affinity for H than P

21. Which one of the following arrangements is the incorrect representation of the property indicated with it ?
- a) $\text{Br} < \text{Cl} < \text{F}$: Electro negativity b) $\text{F} < \text{Br} < \text{Cl}$: Electron - affinity
c) $\text{F}_2 < \text{Br}_2 < \text{Cl}_2$: Bond energy d) $\text{Br}_2 < \text{Cl}_2 < \text{F}_2$: Oxidising strength
22. The C-C single bond length is $1.54 \text{ \AA}$ and that of Cl-Cl is $1.98 \text{ \AA}$. If the electronegativity of Cl and C are 3.0 and 2.5 respectively, the C-Cl bond-length will be equal to
- a) $3.12 \text{ \AA}$ b) $1.67 \text{ \AA}$ c) $1.71 \text{ \AA}$ d) $2.12 \text{ \AA}$
23. Ionisation energy values of an atom are 495, 767, 1250 and $4540 \text{ kJ mole}^{-1}$ the formula of its sulphate is
- a) MSO_4 b) M_2SO_4 c) $\text{M}_2(\text{SO}_4)_3$ d) $\text{M}(\text{SO}_4)_2$
24. If electro negativity of x be 3.2 and that of y be 2.2, the percentage ionic character of xy is
- a) 19.5 b) 18.5 c) 9.5 d) 29.5
25. The electronegativity of carbon from the following data is :
 $E_{\text{H-H}} = 104.2 \text{ kcal mol}^{-1}$, $E_{\text{C-C}} = 83.1 \text{ kcal mol}^{-1}$, $E_{\text{C-H}} = 98.8 \text{ kcal mol}^{-1}$, $X_{\text{H}} = 2.1$
- a) 3.0 b) 2.1 c) 2.5 d) 3.1
26. Which of the following statement is incorrect?
- a) Oxide of aluminium (Al_2O_3) , and arsenic (As_2O_3) are amphoteric.
b) Oxide of chlorine (Cl_2O_7) is less acidic than oxide of nitrogen (N_2O_5).
c) Oxide of carbon (CO_2) is more acidic than oxide of silica (SiO_2).
d) The correct increasing order of basic character of various oxides is $\text{H}_2\text{O} < \text{CuO} < \text{MgO} < \text{CaO}$
27. S-1 : Formation of Mg^{2+} and Al^{3+} , both require the absorption of energy.
 S-2 : The following set of elements represent the correct order of electron affinity values
 $\text{S} > \text{Se} > \text{Te} > \text{O}$
 S-3 : The size of the isoelectronic species is affected by electron-electron interaction in the outer orbitals.
 S-4 : Chemistry of the elements depend on the valence shell electron configurations as well as nuclear masses.
- a) TTFf b) TTFf c) TTfT d) TTTT

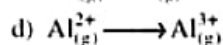
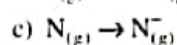
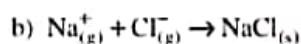
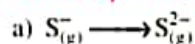
28. $N_0/2$ atoms of X(g) are converted into $\text{X}^+(\text{g})$ by energy E_1 . $N_0/2$ atoms of X(g) are converted into $\text{X}^-(\text{g})$ by the energy E_2 . Hence ionisation potential and electron affinity of X(g) are :
- a) $\frac{2E_1}{N_0}, \frac{2(E_1 - E_2)}{N_0}$ b) $\frac{2E_1}{N_0}, \frac{2E_2}{N_0}$ c) $\frac{(E_1 - E_2)}{N_0}, \frac{2E_2}{N_0}$ d) $\frac{E_1}{N_0}, \frac{2E_2}{N_0}$

More than One correct answer Type Questions

29. The correct statement(s) among the following is (are)
- a) first ionization potential of Al is less than the first ionization potential of Mg
b) second ionization potential of Mg is greater than the second ionization potential of Na
c) first ionization potential of Na is less than the first ionization potential of Mg
d) third ionization potential of Mg is greater than the third ionization potential of Al

30. Which of the following is the incorrect statement
- The second ionization potential of an atom is always less than the first ionization potential
 - The addition of an electron to a neutral atom is an endothermic process
 - Flourine has the maximum electronegativity
 - The size of the cation is always less than the size of the neutral atom
31. Which of the following statements are correct ?
- F is the most electronegative and Cs is the most electropositive available element
 - The electro negativity of halogens decreases from F to I
 - The electron affinity of Cl is higher than that of F though their electro negativity is the reverse
 - The electron affinity of noble gases is low.
32. Which of the following have a diagonal relationship ?
- Li $\rightarrow$ Mg
 - B $\rightarrow$ Mg
 - Be $\rightarrow$ Al
 - Be $\rightarrow$ Na
33. Which of the following sequences is(are) incorrect with respect to the property indicated
- Oxidising power : $F_2 > Cl_2 > Br_2$
 - Bond energy : $F_2 > Cl_2 > Br_2$
 - Electronegativity : $F > Cl > Br$
 - Electron affinity : $F > Cl > Br$
34. Which of the following statement(s) is(are) true?
- ionisation energy $\propto \frac{1}{\text{Screening effect}}$
 - The ionisation energies of Be and Mg are more than ionisation energies of B and Al respectively
 - Atomic and ionic radii of Niobium and Tantalum are almost same
 - Metallic and covalent radii of potassium are $2.3A^0$ and $2.03A^0$ respectively
35. Pick out the correct statement(s) of the following.
- All atoms with an odd atomic number are necessarily paramagnetic.
 - All atoms with an even atomic number are necessarily diamagnetic
 - All atoms with an even atomic number may be diamagnetic or paramagnetic
 - Atoms with an odd atomic number may be paramagnetic and in some cases diamagnetic.
36. Which of the following statement(s) is(are) correct ?
- Fluorine has the highest electro negativity
 - Helium has the highest ionization energy
 - Alkali metals are the strongest oxidizing agents
 - Carbon has the highest melting point among non metals
37. 22.44 kJ energy is required to convert 8 gm of gaseous atom of metal M to $M^+(g)$ if IE_1 of metal M = 374 kJ/mole. Select correct for above metal M
- 0.6 mole gaseous ion (M^+) are formed
 - Same energy can convert all $M^+_{(g)}$ to $M^{2+}_{(g)}$
 - Atomic mass of metal = 133.33
 - 3.6135×10^{22} atoms of M are converted to $M^+_{(g)}$
38. Electron addition would be easier in:
- O
 - O^+
 - O^-
 - O^{2+}

39. Select equations having endothermic step.



40. Following statements regarding the periodic trends of chemical reactivity of the alkali metals and the halogens are given. Which of these statements are incorrect

- The reactivity decreases in the alkali metals but increases in halogens with increase in atomic number down the group
- In both the groups, the chemical reactivity decreases with increase in atomic number down the group
- In both the groups, the chemical reactivity increases with increase in atomic number down the group
- In alkali group, the chemical reactivity increases but decrease in the halogens with increase in atomic number down the group

41. Which of the following statements are correct

- Cl has highest EA among all the known elements
- Both Br and Hg are liquids at room temperature
- Cl is most electronegative element in periodic table
- Atomic radius of noble gases is lowest in their respective periods.

42. Which elements among the following have same electronegativity

- P and H
- S and P
- Cl and N
- S and O

Linked Comprehension Type Questions

Passage-I :

The first ($\Delta_i H_1$) and the second ($\Delta_i H_2$) ionisation enthalpies in kJ mol^{-1} and the electron gain enthalpy ($\Delta_{eg} H$) in kJ mol^{-1} of a few elements are given below :

Elements	$\Delta_i H_1$	$\Delta_i H_2$	$\Delta_{eg} H$
P	520	7300	- 60
Q	419	3051	- 48
R	1681	3374	- 328
S	1008	1846	- 295
T	2372	5251	+ 48
U	738	1451	- 40

Answer the following questions. Only one alternative is correct.

43. Which one of the above elements is least reactive?

- R
- S
- T
- U

44. Which one of the above elements is most reactive metal?

- P
- Q
- U
- S

45. The metal which can form a stable binary halide of the formula MX_2 (X = halogen) :

- P
- Q
- U
- S

Passage-II :

Ionization potential is the minimum amount of energy needed to remove the outermost electron from the gaseous isolated atom. Its unit is eV/atom or kJ/mol. Successive ionization energy is the amount of energy needed to remove electron successively from a gaseous ion, it may be termed as $IE_1, IE_2, \dots$. The difference in the values of $IE_1, IE_2, \dots$ helps to determine electronic configuration of the elements.

Element	IE_1	IE_2	IE_3 (kcal/mol)
P	497	947	1500
Q	98	735	1100
R	176	347	1850
S	296	530	2050

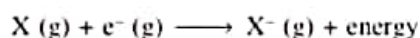
46. Which of the element can make MX type compound (X is the halogen)?
 a) P b) Q c) R d) S
47. The order of ionic mobility in aqueous solution of the following ions will be (If R, S belong to same group):
 a) $R_{(aq)}^{2+} > S_{(aq)}^{2+}$ b) $S_{(aq)}^{2+} > R_{(aq)}^{2+}$ c) $S_{(aq)}^{2+} = R_{(aq)}^{2+}$ d) can not be co-related
48. Of the following four elements which represents an inert gas?
 a) P b) Q c) R d) S

Passage-III :

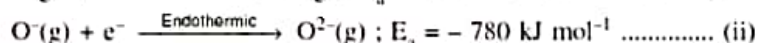
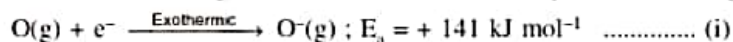
The amount of energy required to remove the most loosely bound electron from an isolated gaseous atom is called as first ionization energy (IE_1). Similarly the amount of energies required to knock out second, third etc. electrons from the isolated gaseous cations are called successive ionization energies and $IE_3 > IE_2 > IE_1$.

(i) Nuclear charge (ii) Atomic size (iii) penetration effect of the electrons (iv) shielding effect of the inner electrons and (v) electronic configurations (exactly half filled & completely filled configurations are extra stable) are the important factors which affect the ionisation energies.

Similarly the amount of energy released when a neutral isolated gaseous atom accepts an extra electron to form gaseous anion is called electron affinity.



A positive electron affinity indicates that the ion X^- has a lower more negative energy than the neutral atom X. The second electron affinity for the addition of a second electron to an initially neutral atom is negative because the electron repulsion out weighs the nuclear attraction, e.g.,



The electron affinity of an element depends upon (i) atomic size (ii) Nuclear charge & (iii) electronic configuration. In general, ionisation energy and electron affinity increases as the atomic radii decrease and nuclear charge increases across a period. In general, in a group, ionisation energy and electron affinity decrease as the atomic size increases.

The members of third period have some higher (e.g. S and Cl) electron affinity values than the members of second period (e.g. O and F) because second period elements have very small atomic size. Hence there is a tendency of electron-electron repulsion, which results in less evolution of energy in the formation of corresponding anion.

49. Identify the least stable ion amongst the following.
- a) Li^- b) Be^- c) B^- d) C^-
50. The first ionisation energy of Na, Mg, Al and Si are in the order of :
- a) $\text{Na} < \text{Mg} > \text{Al} < \text{Si}$ b) $\text{Na} > \text{Mg} > \text{Al} > \text{Si}$
c) $\text{Na} < \text{Mg} < \text{Al} > \text{Si}$ d) $\text{Na} > \text{Mg} > \text{Al} < \text{Si}$
51. Which one of the following statements is incorrect in relation to ionisation enthalpy ?
- a) Ionization enthalpy increases for each successive electron.
b) The greatest increase in ionization enthalpy is experienced on removal of electron from core of noble gas configuration.
c) End of valence electrons is marked by a big jump in ionization enthalpy.
d) Removal of electron from orbitals bearing lower n value is easier than from orbital having higher n value.

Passage-IV :

Atoms of metals have only a few electrons in their valence shells while atoms of non-metals generally have more electrons in their valence shells. Metallic character is closely related to atomic radius and ionisation enthalpy. Metallic character increases from top to bottom in a group and decreases from left to right in a period. Metallic character is inversely related to electronegativity.

52. Considering the elements B, C, N, F and Si, the correct order of their non-metallic character is :
- a) $\text{B} > \text{C} > \text{Si} > \text{N} > \text{F}$ b) $\text{Si} > \text{C} > \text{B} > \text{N} > \text{F}$
c) $\text{F} > \text{N} > \text{C} > \text{B} > \text{Si}$ d) $\text{F} > \text{N} > \text{C} > \text{Si} > \text{B}$
53. The electronegativities of the following elements increase in the order :
- a) C, N, Si, P b) N, Si, C, P
c) Si, P, C, N d) P, Si, N, C
54. The element with maximum electronegativity belongs to :
- a) period 2, group 17 b) period 1, group 18
c) period 3, group 17 d) period 2, group 16

Matrix Matching Type Questions

55. Match the element in Column-I with the periodic property in Column-II.

Column-I

- A) F
B) Cl
C) Fe
D) He

Column-II

- P) Maximum ionization energy
Q) Maximum electronegativity
R) Maximum electron affinity
S) Variable oxidation state

56. Column-I contains the atomic number of few elements and Column-II has the properties of the elements/salts. Accordingly match the Column-I and Column-II.

Column-I

- A) $Z = 19$
 B) $Z = 17$
 C) $Z = 30$
 D) $Z = 20$
 E) $Z = 51$

Column-II

- P) Diamagnetic and has higher atomic radius than $Z = 29$.
 Q) Lower density than either side of the elements in the same group.
 R) Acts as metal as well as non-metal.
 S) Oxide is amphoteric
 T) More negative electron gain enthalpy than $Z = 9$

57. **Column-I (Properties)**

- A) Ionisation energy (IE_1)
 B) Electron affinity (EA_1)
 C) Electronegativity
 D) Electro positive character

Column-II (Magnitude)

- P) Highest in halogens in their respective periods
 Q) Highest in noble gas in their respective periods
 R) Highest in alkali metals in their respective period
 S) Lowest in noble gases in their respective periods

58. Match the values of 1st and 2nd ionisation energy and electron gain enthalpy given in the Column-I with types of elements/compounds given in Column-II.

Column-I

$\Delta_f H_f$, $\Delta_f H_f$, $D_{eg} H$ (in KJ mol^{-1})
 of few elements

- A) 2372 5251 + 48
 B) 419 3051 - 48
 C) 1681 3374 - 333
 D) 1008 1846 - 295

Column-II

- P) Element which exists as a strong reducing agent
 Q) Element which exists as a monoatomic molecule
 R) Least reactive non-metal
 S) Element which acts as a strong oxidising agent
 T) Element which oxide is a stronger basic in nature

Integer Type Questions

59. Common oxidation state of Inner transition elements is ____
 60. Ionisation potential and electron affinity of an element are 17.92 eV and 4.48 eV respectively. The electronegativity of the element on pauling scale is ____
 61. The value of highest Oxidation state exhibited by the element in Modern Periodic Table is ____
 62. Total number of enthalpy(s) (out of given nine) of $A_{(g)}$ which is/are not associated with conversion of $A_{(g)}^- \longrightarrow A_{(g)}^{4+}$
 $IE_1, IE_2, IE_3, IE_4, IE_5, IE_6, EGE_1, EGE_2$ (I.E. - ionization energy, EGE = electron gain enthalpy)
 63. Find the total number of electrons in Cl has Z_{eff} less than 10, when σ is calculated by Slater rules
 64. Write the total number of elements (upto atomic number = 10), which have positive electron gain enthalpy (EGE_1)
 65. Write the approximate EA of atom X in eV/atom unit using the data given below:
 I.E. of X = 13.0 eV/atom ; E.N. of X = 3.05 (on pauling scale)

KEY SHEET (PRACTICE SHEET)

EXERCISE-I

LEVEL-I

- 1) 2 2) 1 3) 2 4) 1 5) 3 6) 3 7) 1 8) 3
 9) 3 10) 2 11) 3 12) 3 13) 2 14) 1 15) 3 16) 1
 17) 1 18) 1 19) 1 20) 2 21) 2 22) 4 23) 1 24) 4
 25) 2 26) 3 27) 1 28) 4 29) 3 30) 1 31) 4 32) 4
 33) 1 34) 3 35) 1 36) 4 37) 2 38) 2 39) 3 40) 4
 41) 3 42) 4 43) 4 44) 4 45) 1 46) 3 47) 2 48) 1
 49) 3 50) 1 51) 1 52) 4 53) 1

LEVEL-II

- 1) c 2) c 3) c 4) c 5) a 6) b 7) a 8) b
 9) d 10) c 11) a 12) b 13) b 14) b 15) c 16) d
 17) c 18) b 19) b 20) a 21) b 22) b 23) c 24) c
 25) d 26) b 27) a 28) d 29) b 30) a 31) d 32) d
 33) c 34) a 35) b 36) a 37) b 38) b 39) bd 40) bc
 41) abcd 42) ab 43) acd 44) abc 45) acd 46) abcd 47) ac 48) acd
 49) abd 50) abc 51) c 52) a 53) c 54) a 55) d 56) c
 57) A-R; B-S; C-P; D-Q 58) A-PQ; B-PQ; C-R; D-RS
 59) A-PQRT; B-PQ; C-S; D-QRT 60) 6 61) 8 62) 4 63) 5
 64) 1 65) 8 66) 8

EXERCISE-II

LEVEL-I

- 1) 3 2) 4 3) 1 4) 2 5) 4 6) 1 7) 3 8) 3
 9) 3 10) 2 11) 3 12) 2 13) 2 14) 4 15) 1 16) 3
 17) 4 18) 3 19) 1 20) 4 21) 2 22) 3 23) 3 24) 4
 25) 3 26) 3 27) 1 28) 3 29) 2 30) 1 31) 1 32) 1
 33) 2 34) 2

LEVEL-II

- 1) d 2) c 3) a 4) c 5) b 6) b 7) c 8) d
 9) c 10) a 11) d 12) d 13) d 14) d 15) a 16) c
 17) c 18) b 19) a 20) c 21) b 22) c 23) c 24) a
 25) c 26) b 27) a 28) b 29) acd 30) abd 31) abc 32) ac
 33) bd 34) abcd 35) ac 36) abd 37) cd 38) d 39) acd 40) abc
 41) ab 42) ac 43) c 44) b 45) c 46) b 47) a 48) a
 49) b 50) a 51) d 52) d 53) c 54) a
 55) A-Q; B-PS; C-S; D-P 56) A-Q; B-T; C-PS; D-P; E-RS
 57) A-Q; B-PS; C-P; D-RS 58) A-Q; B-PT; C-S; D-R
 59) 3 60) 4 61) 8 62) 3 63) 7 64) 4 65) 4

ADDITIONAL PRACTICE EXERCISE

LEVEL-I (MAIN)

1. The common oxidation state exhibited by inner transition elements usually in their compounds is
1) + 2 2) + 3 3) + 5 4) Zero
2. The pair of atomic numbers which represent the p-block elements
1) 6, 12 2) 7, 53 3) 19, 35 4) 38, 51
3. Which of the following is an element present in the d-block, but not a transition element
1) Cd 2) Cu 3) Ca 4) Cr
4. Which of the following is an alloy of non-transition elements
1) Elektron 2) Brass 3) Bronze 4) German silver
5. Common oxidation state of elemental transition metal is
1) +1 2) 0 3) + 3 4) + 2
6. Configuration that does not denote a transition element
1) $3d^1 4s^2$ 2) $3d^{10} 4s^1$ 3) $3d^{10} 4s^2 4p^2$ 4) $3d^8 4s^2$
7. An element of 5f-series but has no electrons filled in 5f-sub shell
1) Ac 2) Ce 3) Th 4) U
8. An element has 18 electrons in the outer most shell. The element is
1) Transition metal 2) Rare earth metal
3) Alkaline earth metal 4) Alkali metal
9. Element with electronic arrangement $[Ar]3d^2 4s^2$ belongs to
1) s-block 2) p-block 3) d-block 4) f-block
10. (A) : The transition metal ions are generally paramagnetic in nature
(R) : Metal ions with incompletely filled d-orbitals are paramagnetic in nature.
1) Both (A) and (R) are true and (R) is the correct explanation of (A)
2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
3) (A) is true but (R) is false
4) (A) is false but (R) is true
11. Number of outer shells partially filled for representative elements
1) Zero 2) One 3) Two 4) Three
12. The atomic number of the element which is not included in the main body of the periodic table
1) 43 2) 57 3) 68 4) 80
13. Zinc is not considered as a transition metal because
1) It is diamagnetic 2) It is not known to form alloys
3) It has no unpaired electrons 4) It has white shade
14. The electronic configuration of an element 'X', is $1s^2 2s^2 2p^6 3s^2 3p^3$. What is the atomic number of the element which is just below 'X' in the periodic table
1) 33 2) 34 3) 31 4) 49

15. What is wrong about transition metals?
 1) They are diamagnetic
 2) They are paramagnetic
 3) They form complexes
 4) They show variable oxidation state
16. Which of the following is a rare earth element?
 1) Cadmium
 2) Californium
 3) Cerium
 4) Cesium
17. Variable valency is exhibited by
 1) Normal elements
 2) Metallic elements
 3) Transitional elements
 4) Non-metallic elements
18. Which one of the following is correct about stability of the given ions ?
 1) $\text{Pb}^{2+} > \text{Pb}^{4+}$
 2) $\text{Pb}^{4+} > \text{Pb}^{2+}$
 3) $\text{Si}^{2+} > \text{Si}^{4+}$
 4) $\text{Sn}^{2+} > \text{Sn}^{4+}$
19. The transition element among the following is
 1) Cu
 2) Sn
 3) Pb
 4) Zn
20. Oxide that is most acidic
 1) Cl_2O_7
 2) SO_3
 3) P_4O_{10}
 4) N_2O_5
21. Generally the nature of the non-metal oxides is
 1) Basic
 2) Acidic
 3) Amphoteric
 4) Neutral
22. The outermost electronic configuration of most electropositive element is
 1) ns^1
 2) $ns^2 np^2$
 3) $ns^2 np^3$
 4) $ns^2 np^5$
23. Chemical similarity between B and Al is due to
 1) Diagonal relationship
 2) Both belong to same period
 3) Similar outer electronic configuration
 4) Inert pair effect
24. Pair of ions with similar ionic radii
 1) $\text{Li}^+, \text{Mg}^{2+}$
 2) Li^+, Na^+
 3) $\text{Mg}^{2+}, \text{Ca}^{2+}$
 4) $\text{Mg}^{2+}, \text{K}^+$
25. Among the following pairs of elements, the pair that is different from others is
 1) Lithium and Magnesium
 2) Nitrogen and Phosphorus
 3) Beryllium and Aluminium
 4) Boron and Silicon

LEVEL-II

LECTURE SHEET (ADVANCED)

More than One correct answer Type Questions

1. Which of the following triads differ in sizes ?
 a) $\text{Na}^+, \text{Mg}^{+2}, \text{Al}^{+3}$
 b) $\text{F}^-, \text{Ne}, \text{O}^{-2}$
 c) Fe, Co, Ni
 d) $\text{Mn}^+, \text{Fe}^{+2}, \text{Cr}$
2. The factors that influence the ionisation energy
 a) the size of the atom
 b) the charge on the nucleus
 c) how effectively the inner shell electrons screen the nuclear charge
 d) type of orbital in which electron is present
3. On moving down the group from F to I, which of the properties decrease ?
 a) Ionic radius
 b) Ionisation energy
 c) Oxidizing property
 d) Electronegativity

4. In which of the following are the correct orders of electron affinity ?
 a) $S > O^-$ b) $O^- < S^{2-}$ c) $O^- > S^{2-}$ d) $N^- > P$
5. The process in which liberation of energy associated in the
 a) $F \rightarrow F^-$ b) $H \rightarrow H^-$ c) $Cl \rightarrow Cl^-$ d) $O \rightarrow O^{2-}$
6. In which of the following arrangements, the order is correct according to the property indicated against it ?
 a) $Al^{3+} < Mg^{2+} < Na^+ < F^-$ – increasing ionic size
 b) $B < C < N < O$ – increasing first ionization enthalpy
 c) $I < Br < F < Cl$ – increasing electron gain enthalpy (with negative sign)
 d) $Li < Na < K < Rb$ – increasing metallic radius
7. Amongst the following statements, which is (are) correct ?
 a) Electro negativity of sulphur is greater than that of oxygen
 b) Electron affinity of oxygen is smaller than that of sulphur
 c) Electron affinity of fluorine is most negative
 d) Electron affinity of chlorine is most negative
8. The molecules having one unpaired electron are :
 a) NO b) CO c) CN^- d) O_2^-

Linked Comprehension Type Questions

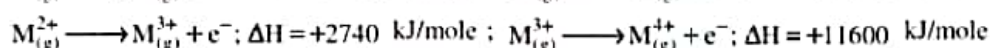
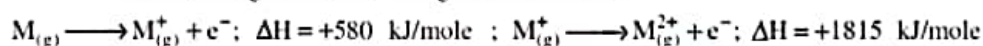
Passage-I :

The amount of energy released when an electron is added to an isolated gaseous atom to produce a monovalent anion is called electron affinity or first electron affinity or electron gain enthalpy. The first electron is given a negative sign as the addition of an electron to a neutral atom is an exoergic process. The addition of electron to A^{-1} requires energy to overcome the force of repulsion. Thus, the second electron affinity is an endoergic process. The magnitude of electron affinity depends on a number of factors such as (i) atomic size (ii) effective nuclear charge (iii) screening effect (iv) half and fully filled orbitals and (v) shape of orbital.

9. Which one of the following arrangements represents the correct order of electron gain enthalpy (with negative sign) of the given atomic species ?
 a) $F < Cl < O < S$ b) $S < O < Cl < F$
 c) $O < S < F < Cl$ d) $Cl < F < S < O$
10. Which of the following species has the highest electron affinity?
 a) F^{-1} b) O^- c) Na d) O
11. Which of the following processes is endoergic in nature ?
 a) $O^- + e^- \rightarrow O^{2-}$ b) $Cl + e^- \rightarrow Cl^-$
 c) $S + e^- \rightarrow S^-$ d) $F + e^- \rightarrow F^-$

Passage-II :

Consider the following ionization energies for a metal M



12. Total no. of electron(s) in outermost shell of metal M are:
 a) Two b) Three c) Four d) One
13. Which one of the four ions has largest electron affinity?
 a) $M_{(g)}^+$ b) $M_{(g)}^{2+}$ c) $M_{(g)}^{3+}$ d) $M_{(g)}^{4+}$
14. Select correct order.
 a) $M_{(g)}^{2+} < M_{(g)}^{4+}$ (size) b) $M_{(g)}^{4+} < M_{(g)}^{3+}$ (ionization energy)
 c) $M_{(g)} < M_{(g)}^+$ (electron affinity) d) $M_{(g)}^{3+} < M_{(g)}^{2+}$ (Z/e ratio)

Integer Type Questions

15. An element 'Y' belong to 1st period and 1st group in Modern Periodic Table. Then atomic number of Y is ____
16. Number of possible ionisation potential values for carbon is ____
17. The IP_1 , IP_2 , IP_3 , IP_4 and IP_5 of an element are 7.1, 14.3, 103.4, 166.8, 262.2 eV respectively. The element belongs to ____ group.

PRACTICE SHEET (ADVANCED)

More than One correct answer Type Questions

1. When an atom of an electronegative element becomes anion, which of the following will not occur?
 a) It acts as a reducing agent b) It loses electrons
 c) Its ionic radius becomes larger d) It accepts electrons
2. On going down a main subgroup in the periodic table (example Li to Cs in IA or Be to Ra in IIA) the expected trend of change in atomic radius is not a/an
 a) Continuous increase b) Continuous decrease
 c) An increase followed by decrease d) A decrease followed by increase
3. Identify the incorrect statement.
 a) Filling of 5d orbital begins with Hf in 5th period
 b) Filling of 4f orbital begins with Ce in 6th period
 c) Filling of 5d orbital begins with La in 5th period
 d) Filling of 4f orbital begins with La in 6th period
4. Identify the incorrect order in which the ionic radius of the following ions increases :
 I) F^- II) Na^+ III) N^{3-}
 a) III, I, II b) I, II, III c) II, III, I d) II, I, III
5. The atomic radius decreases in a period is due to
 a) Increase in nuclear attraction b) Decrease in nuclear attraction
 c) Increase in number of electrons d) Decrease in number of electrons
6. Among the following, the number of elements showing only one non-zero oxidation state is
 a) F b) Na c) Li d) Cl
7. Very slight decrease in atomic radius occurs in a transition series when compared with that in a representative series. This is not due to
 a) Shielding effect b) Penetrating effect
 c) Compton effect d) Inert pair effect

8. If the radius of Fe^{2+} is $0.76\text{\AA}$; the radius of Fe^{3+} will not be
a) $0.64\text{\AA}$ b) $0.76\text{\AA}$ c) $0.88\text{\AA}$ d) $1.08\text{\AA}$
9. Correct statements concerning element with atomic number 10 are
a) forms a covalent network solid b) element is mono atomic
c) has almost zero value of E.A d) has very high value of I.P.
10. Choose the correct statement(s)
a) The maximum positive oxidation state shown by any element is equal to the total number of electrons (s and p) in valence shell
b) The maximum oxidation state shown by elements in a group is also known as group oxidation number
c) Group oxidation number is the most common or most stable oxidation state for a particular element
d) All the elements in a group form some compounds in which they exhibit their group oxidation number.
11. Anomalous pair among the following are
a) Co-Cu b) Ar - K c) Te - I d) Th - Pr
12. Which of the following show amphoteric behaviour ?
a) $\text{Zn}(\text{OH})_2$ b) BeO c) Al_2O_3 d) SO_2
13. The screening effect of inner electrons of an atom causes
a) decrease in ionisation energy b) increase in atomic size
c) increase in ionisation energy d) decrease in atomic size
14. Fluorine is stronger oxidizing agent than chlorine in aqueous solution. This can be attributed to the property:
a) Heat of dissociation b) Electron affinity
c) Ionization potential d) Heat of hydration
15. Diagonal relationship are shown by:
a) Be and Al b) Li and Mg c) Mg and Al d) B and P
16. The number of which subatomic particle is same in case of chlorine atom ^{35}Cl and its chloride ion?
a) Electron b) Proton c) Neutron d) All of these
17. In which of the following pairs size of 1st element is higher as compare to 2nd?
a) Sc, Zn b) Sc, La c) Cu, Ag d) La, Hf

Linked Comprehension Type Questions

Passage-I :

The concept of electronegativity (EN) is mainly to discuss the nature of chemical bonding. EN of an atom is the measure of its capacity to attract the shared electron pair towards itself. Linus Pauling compiled the values of EN that are most widely used, and that are displayed for a few elements. These values are based on bond energy differences between atoms bonded to themselves (e.g. A - A) and to other atoms (e.g. A - B), and an arbitrary value of 4 to fluorine (most electronegative element)

18. Based upon the electronegativity difference of bonded atoms, which of the following is correct about the chemical bond?
- Ionic if $\Delta EN < 1.2$; Polar covalent if $\Delta EN > 1.2$
 - Ionic if $\Delta EN > 1.2$ Polar covalent if $\Delta EN < 1.12$
 - Ionic if $\Delta EN < 2.0$; Polar covalent if $\Delta EN > 2.0$
 - Ionic if $\Delta EN > 2.0$; Polar covalent if $\Delta EN < 2.0$
19. Which of the following best describes the relationship of electronegativity, bond energy and bond moment in a diatomic molecule ?
- As electronegativity increases, both bond energy and bond moment increases
 - As electronegativity increases, bond energy decreases but bond moment increases
 - As electronegativity increases, bond energy increases but bond moment decreases
 - As electronegativity increases, both bond energy and bond moment decreases
20. Element with highest electronegativity is
- Chlorine
 - Fluorine
 - Oxygen
 - Nitrogen

Passage-II :

In general, electron affinity increases as the atomic radii decrease in a period. However, there are exceptions when the atoms have stable configurations. In a group, electron affinity decreases as the size increases. However, the members of 3rd period have somewhat higher values than the members in the 2nd period of the same subgroups.

21. What is the ΔH of the following reaction?
- $$\text{Mg(g)} + 2\text{F(g)} \longrightarrow \text{Mg}^{2+}(\text{g}) + 2\text{F}^{-}(\text{g}) \text{ or } \text{MgF}_2(\text{g})$$
- If electron affinity of $\text{F(g)} = -328 \text{ kJ mol}^{-1}$ and
First ionisation energy of $\text{Mg} = 737.7 \text{ kJ mol}^{-1}$ and
Second ionisation energy of $\text{Mg} = 1451 \text{ kJ mol}^{-1}$
- $1532.7 \text{ kJ mol}^{-1}$
 - $1860.7 \text{ kJ mol}^{-1}$
 - $2516.7 \text{ kJ mol}^{-1}$
 - $2844.7 \text{ kJ mol}^{-1}$
22. The electron affinities of halogens are :
 $\text{F} = -332, \text{Cl} = -349, \text{Br} = -324, \text{I} = -295 \text{ kJ mol}^{-1}$
The higher value of Cl as compared to that of F is due to :
- higher atomic radius of F
 - smaller electronegativity of F
 - weaker electron repulsion in Cl
 - more vacant p-subshell in Cl
23. Which of the following processes is endoergic in nature?
- $\text{O}^{-} + \text{e} \rightarrow \text{O}^{2-}$
 - $\text{Cl} + \text{e} \rightarrow \text{Cl}^{-}$
 - $\text{S} + \text{e} \rightarrow \text{S}^{-}$
 - $\text{F} + \text{e} \rightarrow \text{F}^{-1}$

Matrix Matching Type Questions

24. Match the various decreasing orders listed in column-I with the correct properties listed in column-II

Column-I

- $\text{Fe(III)} > \text{Fe(II)}$
- $\text{Al} > \text{Na}$
- $\text{Cl} > \text{F}$
- $\text{N} > \text{C}$

Column-II

- Electronegativity
- Basic character of their oxides
- Electron gain enthalpy ($-\Delta_{\text{eg}}H$)
- Degree of hydration.

25. Match the groups of species listed in column-I with the characteristic(s)/types of element listed in column-II.

Column-I

- A) SO_2 , NO_3^- , CO_3^{2-}
 B) B, Si, Ge, As, Sb
 C) He, Ne, Ar, Kr, Xe
 D) Na, Mg, Al, Si, P, S

Column-II

- P) Semi-metals
 Q) Isoelectronic species
 R) Vander waal's radii
 S) Typical elements.

Integer Type Questions

26. The period with maximum electronegativity belongs to ____
 27. The group in which all the elements are gases is ____
 28. The atomic number of newly predicted Noble gas element may be $110 + a$. Then the value of 'a' is ____
 29. Nucleus of an element contains 9 protons. Its highest valency would be ____
 30. Which electron gain enthalpy of elements is generally exothermic ____
 31. When the ionisation energy v/s atomic number is plotted for the elements, of atomic number 11 to 18, two peaks are observed for the element X and Y in between the curve. What is the difference between atomic number of element X and element Y?
 32. Count the number of properties, which have higher values for 'O' atom than that of 'S'.
 a) Ionisation potential b) Electron affinity c) Covalent radius
 d) Electronegativity e) (Proton : Electron) ratio

KEY SHEET (ADDITIONAL PRACTICE EXERCISE)**LEVEL-I (MAIN)**

- 1) 2 2) 2 3) 1 4) 1 5) 2 6) 3 7) 3 8) 1 9) 3 10) 1
 11) 2 12) 3 13) 3 14) 1 15) 1 16) 3 17) 3 18) 1 19) 1 20) 1
 21) 2 22) 1 23) 3 24) 1 25) 2

LEVEL-II**LECTURE SHEET (ADVANCED)**

- 1) abd 2) acd 3) bcd 4) ab 5) abc 6) acd 7) bd 8) ad 9) c 10) d
 11) a 12) b 13) d 14) c 15) 1 16) 6 17) 2

PRACTICE SHEET (ADVANCED)

- 1) abd 2) bcd 3) acd 4) abc 5) ac 6) abc 7) bcd 8) bcd 9) bcd 10) ab
 11) bcd 12) abc 13) ab 14) ad 15) ab 16) bc 17) ad 18) d 19) a 20) b
 21) a 22) c 23) a 24) A-PQRS; B-PRS; C-R; D-P 25) A-Q; B-P; C-R; D-S
 26) 2 27) 0 28) 8 29) 1 30) 1 31) 3 32) 2